ADVANCED GRAPH-BASED AND AUTONOMOUS ARCHITECTURES FOR HYPERSPECTRAL ANOMALY DETECTION

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By

VENKATA AKHILA CHIMATA 228W1A12E4

SWATHI GUDIWADA 228W1A12F1

Under the Guidance of

Dr. M. Suneetha, Ph.D,

Professor and Head

DEPARTMENT OF INFORMATION TECHNOLOGY

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(An ISO certified and NBA accredited institution)
Kanuru, Vijayawada – 520007

CERTIFICATE

This is to certify that this project report titled **"Advanced Graph-Based And Autonomous architectures For Hyperspectral Anomaly Detection"** is a bonafide record of work done by **Ms. Venkata Akhila Chimata (228W1A12E4) and Ms. Swathi Gudiwada (228W1A12F1)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

(Dr. M. Suneetha)
Dean R&D,Professor &Head
Dept. of Information Technology

(Dr. M. Suneetha)
Dean R&D, Professor &Head
Dept. of Information Technology

EXTERNAL EXAMINER SIGNATURE

Date of examination:

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R.SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**Advanced Graph-Based And Autonomous Architectures For Hyperspectral Anomaly Detection**
2	**Student Names & Numbers**	**Ms.Venkata Akhila Chimata (228W1A12E4) Ms. Swathi Gudiwada (228W1A12F1)**
3	**Name of The Guide**	**Dr. M. Suneetha**
4	**Research Group**	**Remote Sensing and Deep Learning**
5	**Application Area**	**Airspace Surveillance and Defense Applications**
6	**Aim of the Project**	**1.** To design and implement advanced deep-learning architectures, specifically Gated Transformers and Autonomous Autoencoders, to model complex spectral-spatial correlations in high-dimensional hyperspectral imagery. **2.** To evaluate the models' performance across diverse operational environments (urban, coastal, and island) to ensure consistent reliability in wide-area airspace monitoring
7	**Project Outcomes**	**1.** Provides a high-fidelity modeling framework that accurately distinguishes subtle aircraft signatures from complex geographical and atmospheric backgrounds. **2.** Validates the system's cross-terrain resilience, proving that high-altitude anomalies are successfully identified regardless of the underlying terrestrial complexity.

Student Signatures

1.

2.

Signature of the Guide

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ABSTRACT

The rapid proliferation of advanced aviation and high-altitude propulsion systems has necessitated the development of sophisticated surveillance technologies for national defense. Hyperspectral Anomaly Detection (HAD) stands at the forefront of this evolution, offering the ability to distinguish unknown aircraft, such as high-speed jets and stealth drones, by analyzing their unique spectral fingerprints across the electromagnetic spectrum. However, the operational deployment of HAD is hindered by the extreme complexity of hyperspectral data, characterized by high spectral dimensionality, nonlinear atmospheric interference, and the intense variability of terrestrial backgrounds.In defense scenarios, the primary challenge lies in the "pixel mixing" phenomenon at target boundaries and the subtle spectral differences between high-altitude anomalies and their surrounding environment, which often leads to high false-alarm rates

To overcome these barriers, this research implements a dual-architectural framework that leverages the strengths of diverse deep-learning paradigms. The first module, a Gated Transformer-based Hyperspectral Anomaly Detection (GT-HAD) network, introduces a novel Gated Transformer Block (GTB). This architecture achieves spectral-spatial separation through an Anomaly-Focused Branch (AFB), which suppresses outlier signatures via feature redefinition, and a Background-Focused Branch (BFB), which utilizes patch-level self-attention to enhance background consistency. A Content-Matching Method (CMM) dynamically regulates these branches to ensure optimal reconstruction. The second module, an Autonomous Reconstruction Network (Auto-HAD), employs a Deep Image Prior (DIP) strategy for self-supervised learning. By incorporating an adaptive weighted loss mechanism, the model autonomously prioritizes the reconstruction of dominant background pixels, effectively "masking out" anomalous targets through high reconstruction residuals. This framework provides high-contrast anomaly maps and superior Area Under the Curve (AUC) performance.

Keywords: Hyperspectral Imaging (HSI); Anomaly Detection (HAD); National Defense & Airspace Surveillance; Gated Transformers; Autonomous Reconstruction; Deep Image Prior (DIP); Remote Sensing; Spectral-Spatial Feature Extraction; Self-Supervised Learning; Target Detection

CHAPTER-1

INTRODUCTION

This section outlines the motivation, technical foundation, and objectives of a project that employs state-of-the-art deep learning architectures—specifically the Graph Transformer for Hyperspectral Anomaly Detection (GT-HAD) and the Autonomous Deep Image Prior (Auto-AD)—to analyze and identify rare anomalous targets within complex hyperspectral environments. The research utilizes multi-temporal and high-dimensional spectral data across several benchmark datasets, including Los-Angeles, Gulfport, Texas-Goast, Cat-Island, and Pavia, to evaluate the effectiveness of local attention-based learning versus global structural priors in the field of remote sensing and geospatial analysis. Unlike traditional multispectral systems, the project leverages the full spectral continuum to distinguish sub-pixel anomalies from intricate background clutter, providing a robust framework for unsupervised target detection without prior spectral knowledge.

1.1 Origin of the Problem

Rapid advancements in sensor technology have led to the proliferation of hyperspectral imaging (HSI), which captures data in hundreds of narrow spectral bands. This wealth of information has theoretically made it easier to identify rare materials; however, the "curse of dimensionality" and the non-linear mixing of materials have created significant detection challenges [8]. Traditionally, anomaly detection was performed using statistical methods such as the Reed-Xiaoli (RX) detector, which relied on the assumption that the background follows a multivariate Gaussian distribution. In real-world urban and coastal environments, however, the background is often highly non-stationary and composed of multiple distinct materials, leading to high false-alarm rates and poor reliability in complex terrains [7].

While earlier work using basic Autoencoders (AEs) and Generative Adversarial Networks (GANs) effectively captured some background features, these methods were often limited by the "over-learning" problem. Standard deep learning models tend to be so powerful that they accidentally reconstruct the rare anomalies along with the background, effectively "erasing" the very targets they are supposed to find. Furthermore, traditional spatial-spectral methods lacked the ability to model long-range spectral dependencies and high-order spatial correlations

simultaneously. To address these critical limitations, the present project employs a dual-pipeline comparative framework. We utilize the GT-HAD model, which integrates a Cluster Matching Mechanism (CMM) with Graph Transformers to "gate" the learning process, alongside the Auto-AD model, which uses the internal architectural bias of Skip-Nets to naturally favoru background reconstruction. By comparing these two distinct paradigms, this project aims to establish a more accurate and reliable standard for hyperspectral anomaly detection in diverse geographical scenarios [14].

1.2 Basic Definitions and Background:

1.2.1 Hyperspectral Remote Sensing

Hyperspectral Remote Sensing involves the acquisition of hundreds of contiguous spectral bands across the electromagnetic spectrum, ranging from the visible to the short-wave infrared regions. This results in a 3D data cube where each pixel contains a continuous spectral signature, often called a "spectral fingerprint." This project utilizes high-dimensional datasets from various sensors, including AVIRIS and ROSIS, to map and detect anomalies at fine spectral resolutions [13]. The primary challenge in HSI background modeling is the inherent spectral mixing, where a single pixel may contain contributions from multiple materials like vegetation, soil, and man-made structures.

1.2.2 Hyperspectral Anomaly Detection (HAD)

Hyperspectral Anomaly Detection is an unsupervised task that aims to identify pixels in an HSI cube that deviate significantly from their surrounding background. Unlike target detection, which requires a pre-defined library of spectral signatures, HAD operates blindly. An anomaly is defined as a rare target that occupies only a few pixels and is spectrally distinct from the dominant background. The detection process involves suppressing the background clutter to highlight these rare residuals, typically measured via reconstruction error or statistical distance[18].

1.2.3 Graph Transformers (GT-HAD)

The Graph Transformer for Hyperspectral Anomaly Detection (GT-HAD) is a modern architecture that treats spectral bands as nodes in a graph. It utilizes the self-attention mechanism to learn the relationships between different spectral signatures. A key component is the Cluster Matching Mechanism (CMM), which dynamically identifies whether a local block belongs to a

known background cluster. This project implements the CMM to guide a gated attention system, ensuring the model only learns "Normal Background" features, thereby preventing the reconstruction of anomalous targets.

1.2.4 Autonomous Anomaly Detection (Auto-AD)

Auto-AD represents a paradigm shift in detection, utilizing the "Deep Image Prior" (DIP) concept. It assumes that a neural network's architecture inherently favors the reconstruction of natural, structured patterns (the background) over unstructured or rare patterns (the anomalies). This project employs a Skip-Net (Hourglass) architecture that maps a random noise tensor to the hyperspectral space [16]. By combining this with a Dynamic Residual Weighting loop, the model iteratively "mask out" suspected anomalies during training, ensuring that the final reconstruction is a pure represention of the background clutter.

1.3 Problem Statement

Existing hyperspectral monitoring systems often struggle with the trade-off between background reconstruction precision and anomaly preservation. Conventional methods fail to deliver reliable detection in "busy" environments where the background is multi-modal or where anomalies are sub-pixel in size. The absence of robust spatio-temporal modeling and the tendency of deep networks to over-fit to anomalies results in a high number of false positives and missed targets [15]. To overcome these limitations, this project proposes a comparative evaluation of GT-HAD and Auto-AD. By integrating advanced local gating (CMM) and global structural priors, the project seeks to determine the optimal balance for accurately modeling historical spectral trends and predicting the location of rare targets in diverse terrain types, thereby supporting more reliable and automated geospatial surveillance strategies.

1.3.1 Objectives

1. To implement and optimize the GT-HAD architecture, specifically tuning the Cluster Matching Mechanism (CMM) to handle varying levels of background complexity in datasets like Pavia and Los-Angeles.

2. To deploy the Auto-AD framework using a Deep Image Prior approach, evaluating its ability to reconstruct broad background areas without the need for pre-training or external libraries.

3. To conduct a rigorous comparative analysis using the AUC-ROC metric to benchmark the

performance of both models across six distinct hyperspectral datasets, establishing a performance baseline for transformer-based HAD.

4. To analyze the computational trade-offs between local block-wise processing (GT-HAD) and global image reconstruction (Auto-AD) in terms of training time and detection sensitivity.

1.3.2 Outcomes

1. Comprehensive Performance Reports: Provides detailed ROC curves and AUC scores for both models, forming a solid technical foundation for choosing a detection strategy based on terrain type.

2. High-Fidelity Anomaly Maps: Generates precise detection maps that clearly highlight anomalies while successfully suppressed complex background clutter in urban, coastal, and rural scenes.

3. Methodological Insights: Offers a deep understanding of how local attention mechanisms and global architectural priors differ in their handling of sub-pixel spectral mixing.

1.4 Application

1.4.1 Hyperspectral Data Benchmarking and Model Evaluation

The proposed framework plays a critical role in benchmarking advanced anomaly detection models such as GT-HAD and Auto-AD across diverse hyperspectral datasets like Pavia, Gulfport, and Cat-Island. By generating consistent anomaly maps and ROC-based performance metrics, the system enables researchers to evaluate the effectiveness of local attention mechanisms versus global reconstruction priors [11]. This application is essential for establishing standardized evaluation protocols in hyperspectral anomaly detection research**.**

1.4.2 Sub-Pixel Anomaly Detection in High-Dimensional Data

The project is specifically designed to detect anomalies that occupy only a fraction of a pixel, which is a major challenge in hyperspectral imaging. By leveraging full spectral signatures and reconstruction-based residual analysis, the framework can identify subtle spectral deviations that are not visible through conventional methods. This is particularly useful in scenarios where anomalies are deeply embedded within complex background mixtures. This capability significantly improves detection sensitivity and reduces the likelihood of missing critical targets

in real-world applications.

1.4.3 Comparative Analysis of Deep Learning Paradigms

A key application of this work is the direct comparison between transformer-based architectures (GT-HAD) and Deep Image Prior-based models (Auto-AD). The framework allows systematic analysis of how different learning strategies—explicit background gating versus implicit architectural bias—affect anomaly detection performance. This contributes to the development of more robust and generalizable HAD models.

1.4.4 Background Suppression in Complex Spectral Environments

The system is highly effective in suppressing heterogeneous and non-stationary background clutter, which is a core challenge in hyperspectral analysis. By accurately modeling background distributions using CMM-guided attention and DIP-based reconstruction, the framework ensures that anomalies remain prominent in the residual space. This improves detection reliability in urban, coastal, and mixed terrain datasets.

1.4.5 Automated Unsupervised Anomaly Detection Framework

The proposed approach eliminates the need for labeled data or prior spectral libraries, making it suitable for fully unsupervised anomaly detection tasks. This is particularly valuable in real-world hyperspectral scenarios where ground truth data is scarce or unavailable. The system can autonomously learn background characteristics and identify deviations without human intervention.

1.4.6 Residual Map Generation for Anomaly Localization

The project generates high-quality residual maps that highlight anomalous regions based on reconstruction error. These maps serve as a direct visual and quantitative tool for identifying rare targets within hyperspectral scenes. Combined with ROC curve analysis, they provide both interpretability and performance validation for anomaly detection models.

The introduction establishes a strong foundation for addressing the critical challenges associated with hyperspectral anomaly detection, particularly the issues of high spectral dimensionality, non-stationary background distributions, and the difficulty of identifying sub-pixel anomalies in

complex environments. By integrating two advanced and fundamentally different deep learning paradigms—Graph Transformer-based GT-HAD and Deep Image Prior-based Auto-AD—the project creates a comprehensive framework for analyzing both local spectral dependencies and global structural reconstruction behavior.

Furthermore, the study emphasizes the importance of controlled background learning, where GT-HAD utilizes the Cluster Matching Mechanism (CMM) to selectively guide attention toward normal background patterns, while Auto-AD leverages architectural bias to inherently suppress anomalous structures during reconstruction. This dual approach not only addresses the limitations of traditional statistical and deep learning methods but also provides deeper insight into how different modeling strategies impact anomaly preservation and detection accuracy [19].The comparative nature of the framework enables systematic evaluation across multiple benchmark datasets, ensuring that the findings are robust, scalable, and applicable to diverse hyperspectral scenarios. By focusing on residual-based anomaly extraction and ROC-driven performance validation, the project ensures both quantitative rigor and interpretability of results.

In addition, the work highlights the significance of unsupervised learning in hyperspectral analysis, eliminating the dependency on labeled datasets and making the approach adaptable to real-world conditions where prior knowledge is limited or unavailable. The ability to generate high-fidelity anomaly maps while effectively suppressing complex background clutter demonstrates the practical strength of the proposed methodologies.Overall, this research contributes to the advancement of autonomous hyperspectral analysis by bridging the gap between transformer-based attention mechanisms and reconstruction-driven priors. It lays a solid groundwork for future exploration into hybrid models, improved gating strategies, and more efficient computational designs. Furthermore, the proposed framework demonstrates strong adaptability across varying scene complexities by effectively balancing sensitivity to anomalies with robustness against background variations. The integration of transformer-based attention in GT-HAD enables enhanced contextual understanding, while the reconstruction-driven nature of Auto-AD ensures stable background modelling even in highly heterogeneous environments. This complementary behaviour highlights the potential of combining advanced architectural strategies to address the inherent challenges of hyperspectral anomaly detection. Additionally, the scalability of the approach allows it to be extended to larger datasets and complex conditions.

CHAPTER-2

Review of Literature

This literature survey examines existing hyperspectral anomaly detection (HAD) methods, highlighting limitations in traditional statistical, representation-based, and deep learning approaches, and reviews recent advancements in transformer, autoencoder, and tensor-based models for improved background suppression and anomaly separability.

2.1Description of Existing Systems

[1] Synergistic Kolmogorov–Arnold Networks and Fidelity-Gated Transformer for Hyperspectral Anomaly Detection

Authors: Jijun Xiang, Tao Wang, Pengxiang Wang, Cheng Chen, Nian Wang, Jiping Cao and Qiying Wang

Year of Publishing: 2025

Observations: The base paper highlights that traditional hyperspectral anomaly detection models suffer from uniform processing and over-reconstruction issues, which limit their ability to distinguish subtle anomalies from complex backgrounds. It demonstrates that introducing a gating mechanism with a dual-branch architecture significantly improves performance by separating background and anomaly processing based on reconstruction fidelity. This directly aligns with our project, where the GT-HAD model uses a similar gated, branch-based approach, enabling better spectral–spatial feature learning and stronger background suppression, resulting in consistently higher AUC scores across datasets. In contrast, the Auto-HAD model, being purely reconstruction-based, tends to reconstruct anomalies along with the background, leading to performance variability. Thus, the observations from the paper validate that controlled learning through gating is more effective than traditional reconstruction-only methods for accurate hyperspectral anomaly detection.

[2] Enhancing Hyperspectral Anomaly Detection with a Novel Differential Network Approach for Precision and Robust Background Suppression

Authors: Delgado, E., Barboza, E., Rojas-Briceño, N.B. et al.

Year of Publishing: 2024

Observations: It introduces differential convolution with dual receptive-field kernels to extract local punctured neighborhood features; proposes local detail attention and local Transformer attention to enhance inner-window features before differential subtraction; reconstructs clean background directly from differential features without background-constraint bias; outperforms eight state-of-the-art methods on five real-world HSI datasets in both subjective and objective.

[3] Robust multi-stage progressive autoencoder for hyperspectral anomaly detection
Authors: Qing Guo, Yi Cen, Lifu Zhang, Yan Zhang, Shunshi Hu, Xue Liu

Year of Publishing: 2024

Observations: It proposes RMSAD, a multi-stage progressive convolutional autoencoder framework that incrementally learns global distribution and long-range dependencies while preserving local details; introduces Cross-stage Multi-scale Fusion (CMF) and Cross-Stage Spatial Feature Extraction (CSFE) modules to enhance background representation and suppress anomaly reconstruction bias; superior detection accuracy on six HSI datasets compared to state-of-the-art methods; effectively addresses boundary characterization and generalization limitations of conventional AE-based HAD.

verified by ablation studies on both synthetic and real hyperspectral datasets.

[4] Hyperspectral Anamoly Detection with Auto-Encoder and Independent Target

Authors: Shuhan Chen, Xiaorun Li and Yunfeng Yan

Year of Publishing: 2023

Observations: It proposes AE-IT, a novel LRaSMD method using AE encoder weight matrix to construct low-rank background subspace with adaptive latent-layer neurons; projects data into encoder orthogonal subspace, spheres it, and extracts unsupervised independent targets to build anomaly space; applies sparse cardinality (SC) constraint on anomaly component to mitigate noise; competitive performance against LRaSMD-based and AE-based approaches through combined Mahalanobis distance and multi-component detector.

[5] Background Reconstruction via 3D-Transformer Network for Hyperspectral Anomaly Detection

Authors: Ziyu Wu and Bin Wang

Year of Publishing: 2023

Observations: It proposes 3DTR network combining spatial and spectral Transformer modules in series to reconstruct background by capturing spatial correlations among pixels and global spectral similarity among bands; incorporates coarse pre-detection and novel similarity-based patch-generation method to minimize anomaly and irrelevant-pixel contamination; achieves fewer false alarms than traditional and state-of-the-art (model-based and AE-based) detectors;

[6] Hyperspectral Anomaly Detection Based on Adaptive Low-Rank Transformed Tensor

Authors: Bairwa, B., Sharma, R., Kundu, A., Sammen, S.S., Alshehri, F., Pande, C.B., Orban, Z., Salem, A.

Year of Publishing: 2023

Observations: It proposes ALRTT that decomposes HSI into background, anomaly, and noise tensors using adaptive low-rank transformed tensor with data-dependent transform; imposes nuclear norm on frontal slices of the transformed tensor and l_{2,1}-norm on anomaly tensor for group sparsity; adaptively learns low-rank matrix via l_{2,1}-norm minimization of an initialized matrix; proximal alternating minimization algorithm converges to critical point and demonstrates superiority over state-of-the-art methods on four widely used datasets.

2.2 Summary of Literature Study

The Table 2.1 summarizes a comprehensive review of recent research works in the domain of Hyperspectral Anomaly Detection (HAD), with a focus on advanced deep learning approaches such as transformer-based models and reconstruction-based architectures. The literature highlights key trends including the shift from traditional statistical methods to deep neural networks for improved spectral–spatial feature extraction. Important observations include the effectiveness of dual-branch and gating mechanisms in enhancing background suppression. In addition, several studies emphasize the role of self-supervised learning in reducing dependency on labeled data while maintaining high detection performance. Recent advancements also demonstrate the integration of attention mechanisms to better capture long-range spectral–spatial dependencies and global spatial relationships. The summary of the literature survey is presented below in Table 2.1.

Table 2.1: Literature Survey

Title	Year	Algorithms used	Observations
Synergistic Kolmogorov–Arnold Networks and Fidelity-Gated Transformer for Hyperspectral Anomaly Detection [1]	2025	Fidelity-Gated Transformer (GCAT), LARB/GBRB branches, KAN-MLP (Kolmogorov-Arnold Network).	• **Fidelity-Based Gating Impact**: The introduction of the Contextual Feature Matching Module (CFMM) serves as a critical decision point that routes features based on their reconstruction accuracy, effectively solving the "identity mapping" problem where standard transformers accidentally reconstruct anomalies too well. • **Nonlinear Spectral modeling**: By utilizing KAN-MLP with learnable spline activations instead of static ReLU functions, the model successfully captures high-order spectral fluctuations that are characteristic of man-made aircraft materials, which are often "lost" in traditional linear processing. • **Empirical Excellence**: Experimental results show that KANGT maintains near-perfect AUC scores (0.99+) even in "stress-test" scenarios like the Pavia dataset, where anomalies are highly mixed with background pixels.
Enhancing hyperspectral anomaly detection with a novel differential network approach for precision and robust background suppression [2]	2024	DifferNet, Differential Convolution (5x5 & 3x3), Local Detail Attention (LDA), Local Transformer Attention (LTA).	• **Local Punctured Neighborhoods**: The core observation is that anomalies are outliers *locally*; by using nested kernels (5x5 and 3x3), the model extracts features from the "ring" of pixels surrounding a target without including the target itself, creating a perfect reference for reconstruction. • **Optimization Strategy shift**: Unlike traditional models that aim for a "clean background" directly, DifferNet optimizes to reconstruct the original input through differential features, which effectively circumvents the "background constraint bias" that often leads to false positives on terrestrial edges. This allows the model to better.

Robust multi-stage progressive autoencoder for hyperspectral anomaly detection [3]	2024	**RMSAD**, Multi-stage learning, Cross-stage Multi-scale Fusion (CMF), Cross-Stage Spatial Feature Extraction (CSFE).	• **Cascaded Refinement Logic**: The multi-stage architecture observed in this paper allows the network to learn background statistics at multiple scales. Coarse features are refined in early stages, while fine-grained spatial boundaries of anomalies are isolated in the final stages. • **Cross-Stage Information Flow**: The Cross-stage Multi-scale Fusion (CMF) module acts as a "spectral filter" that adaptively weights different HSI bands, ensuring that noisy or redundant bands do not contaminate the latent representation. • **Anomaly Propagation**: Through the Cross-Stage Spatial Feature Extraction (CSFE) module, the model explicitly identifies abnormal spatial patterns and "pushes" them into higher-error residuals.
Hyperspectral anomaly detection with auto-encoder and independent target [4]	2023	AE-IT, LRaSMD (Low Rand and Sparse Decomposition), ATGP target extraction, Sparse Cardinality (SC).	• **Dimensionality Decoupling**: The serial configuration of Spatial and Spectral Transformers allows the model to first solve the "where" (spatial arrangement) and then the "what" (spectral composition), capturing the unique 3D structure of HSI data more effectively than standard 2D convolutions. • **Intelligent Patch Sampling**: The novel similarity-based patch generation strategy selects the top N_p pixels that are spectrally similar to the center pixel. This ensures that the transformer is only "looking at" relevant background pixels during training, drastically reducing contamination from anomalous targets. • **Training Set Purification**: By incorporating a coarse pre-detection step (using RX), the model effectively removes potential anomalies from its training window, forcing the transformer to become an expert only in background reconstruction.

Background Reconstruction via 3D-Transformer Network for Hyperspectral Anomaly Detection [5]	2023	3DTR, Serial Spatial and Spectral Transformers, Similarity-based patch generation, Coarse pre-detection.	• **Subspace Decomposition Mastery**: The model combines the mathematical precision of Low-Rank Sparse Matrix Decomposition (LRaSMD) with the feature extraction power of AEs. It uses the encoder's weight matrix to build an orthogonal projector that separates background and anomaly subspaces. • **Adaptive Structural Tuning**: The observation that latent layer size must match the Virtual Dimension (VD) of the HSI data leads to a more scientific way of preventing "unbiased reconstruction," ensuring deep pixels are only reconstructed if they fit the dominant background distribution. • **Target Saliency**: The Independent Target (IT) extraction mechanism uses an automatic process (ATGP) to find representative anomaly signatures, creating a specific "anomaly reference space" that increases the specificity of the detector.
Hyperspectral anomaly detection based on adaptive low-rank transformed tensor [6]	2023	ALRTT, Tensor-based decomposition, $l_{2,1}$-norm matrix optimization, Proximal Alternating Minimization (PAM).	• **Tensor Integrity preservation**: This paper emphasizes that HSIs are 3D tensors, not 2D matrices. By using t-product-based modeling, the model preserves the high-order correlations between spectra that are normally destroyed when "unfolding" the data into a matrix. • **Group Sparsity Constraint**: The use of the $l_{2,1.1}$-norm on the anomaly tensor allows the model to detect anomalies as connected groups of pixels, which is far more effective for identifying actual aircraft or targets compared to "point-based" detection. • **Mathematical Convergence**: The Proximal Alternating Minimization (PAM) algorithm is proven to converge to critical points, ensuring that the results are stable and consistent across different datasets like Cuprite and SpecTIR.

2.3 Software Requirement Specification

2.3.1 Purpose:

The purpose of this project is to design, develop, and implement an advanced hyperspectral anomaly detection system using state-of-the-art deep learning architectures, namely the Graph Transformer-based Hyperspectral Anomaly Detection (GT-HAD) model and the Autonomous Deep Image Prior-based model (Auto-HAD). The system aims to accurately identify rare and subtle anomalies present in hyperspectral imagery without relying on prior spectral knowledge or labeled datasets.

Unlike traditional statistical approaches such as RX detector or conventional machine learning models, this system leverages deep spectral–spatial feature extraction, gated attention mechanisms, and reconstruction-based learning to effectively separate anomalous targets from complex background environments [20]. The primary objective is to achieve high detection accuracy, strong background suppression, and reliable anomaly localization, particularly in challenging scenarios where anomalies occupy only a small fraction of the image.

2.3.2 Scope:

The scope of the system includes end-to-end processing of hyperspectral datasets obtained from benchmark repositories such as AVIRIS and ROSIS. The system handles multiple datasets including Los Angeles (1 & 2), Gulfport, Texas Coast, Cat Island, and Pavia, each representing diverse geographical and environmental conditions. The system processes high-dimensional hyperspectral imagery (HSI) datasets (Los Angeles, Gulfport, Pavia, etc.) provided in .mat format. It partitions 3D hyperspectral cubes to capture local spatial-spectral contexts and employs dual-branch gating mechanisms and adaptive weighted loss functions to differentiate between dominant background pixels and rare anomalous targets [17]. The scope includes end-to-end functionality from data normalization and patch extraction to deep network optimization, residual diffusion post-processing, and quantitative performance validation using Area Under the Curve (AUC) metrics.

The project encompasses the following stages:

- Data acquisition and preprocessing of hyperspectral cubes
- Spectral normalization and feature preparation

- Implementation of dual deep learning models (GT-HAD and Auto-HAD)
- Background reconstruction and anomaly isolation
- Residual map generation for anomaly detection
- Performance evaluation using quantitative metrics such as AUC-ROC
- Comparative analysis across multiple datasets

The system is designed to handle high-dimensional spectral data and perform unsupervised anomaly detection, making it suitable for real-world applications where labeled data is limited or unavailable.

2.3.3 Intended Users

This specialized earth observation framework is built for national defense researchers and airspace surveillance experts monitoring for unauthorized incursions, remote sensing scientists exploring advanced target-to-background separation techniques, and security agencies requiring automated early-warning systems. It also serves academic researchers in the fields of deep learning and geospatial intelligence, providing a scalable and high-precision tool for identifying spectral deviations in complex terrestrial and maritime environments.

This system is intended for a wide range of users including:

- Remote sensing researchers analyzing hyperspectral data
- Defense and surveillance agencies for detecting hidden or camouflaged targets
- Environmental monitoring organizations tracking rare pollution or hazardous events
- Academic institutions and students studying deep learning and hyperspectral analysis
- Machine learning engineers and developers working on anomaly detection systems

The system provides an advanced platform for analyzing complex hyperspectral datasets and extracting meaningful insights through automated detection mechanisms.

2.3.4 Functional Requirement

1. The system shall load hyperspectral datasets in the form of spectral cubes (.mat files) along with corresponding ground truth maps.

2. It shall preprocess the data by normalizing spectral values and organizing them into suitable input formats for deep learning models.
3. The system shall implement the GT-HAD architecture, which processes data using local patch-based transformers and gating mechanisms to separate anomaly and background features.
4. It shall implement the Auto-HAD model, which reconstructs the hyperspectral image from a noise tensor using a skip-network architecture.
5. The system shall perform background reconstruction for both models, ensuring minimal contamination from anomalous pixels.
6. It shall compute pixel-wise residuals by subtracting reconstructed outputs from original inputs.
7. The system shall generate residual maps and anomaly detection maps, highlighting regions with high spectral deviation.
8. It shall calculate evaluation metrics such as:
    - AUC (Area Under Curve)
    - ROC Curves
    - Detection accuracy and performance comparison
9. The system shall support visualization of results, including anomaly maps and performance graphs.
10. It shall enable comparative analysis between GT-HAD and Auto-HAD models across different datasets.

2.3.5 Non-Functional Requirements

Performance: The system should efficiently process high-dimensional hyperspectral data and complete training within a reasonable time using GPU acceleration.

Accuracy: The models should achieve high anomaly detection performance, ideally with AUC values approaching 1.0, indicating near-perfect classification.

Scalability and Reliability: The framework should support additional hyperspectral datasets with varying spectral bands and spatial dimensions without major modifications.

Usability: The implementation should be user-friendly, with well-structured code, as well as a clear documentation, and modular design for ease of understanding and reuse.

Realibility : The system should produce consistent and stable results across multiple runs and datasets.

Reproducbility: Experimental results should be reproducible using fixed configurations, random seeds, and consistent preprocessing steps.

Portability: The system should be deployable on platforms such as Google Colab, Kaggle, or local machines with GPU support**.**

2.3.6 Software and Hardware Requirements

Platform

The implementation of the proposed hyperspectral anomaly detection system is carried out using a robust and scalable computational platform capable of handling high-dimensional spectral data and complex deep learning architectures. Unlike traditional remote sensing workflows that rely on cloud-based geospatial platforms, this system operates within a deep learning-driven environment where efficient data handling, model training, and evaluation are critical. The entire workflow is designed to support large hyperspectral datasets while ensuring computational efficiency and reproducibility.

The implementation and experimentation of the proposed hyperspectral anomaly detection models are carried out using the Kaggle platform, which provides a reliable and high-performance cloud-based environment for data science and deep learning tasks. Kaggle offers free access to GPU-enabled notebooks, enabling efficient training and evaluation of computationally intensive models such as GT-HAD and Auto-HAD without the need for dedicated local hardware. The platform supports Python-based development with pre-installed libraries like PyTorch, NumPy, and Scikit-learn, ensuring seamless integration of the required tools for hyperspectral data processing and model implementation. Additionally, Kaggle facilitates easy dataset management, allowing hyperspectral datasets to be uploaded, accessed, and processed directly within the notebook environment. Its version control, notebook sharing,

and reproducibility features further enhance collaborative development and experimentation. Overall, the use of Kaggle as the primary platform ensures scalability, accessibility, and efficient resource utilization for implementing and validating the proposed anomaly detection framework.

Coding Environment

The system is developed using Python, which is widely recognized for its flexibility and extensive ecosystem in scientific computing and machine learning. Python provides seamless integration with deep learning frameworks and supports efficient handling of multidimensional hyperspectral data. Its rich set of libraries enables implementation of advanced algorithms, numerical operations, and visualization techniques required for hyperspectral anomaly detection tasks.

Libraries and Frameworks

The core implementation of the system utilizes PyTorch, a powerful deep learning framework that supports dynamic computation graphs and efficient GPU acceleration. PyTorch enables the development of complex architectures such as the Graph Transformer-based GT-HAD model and the reconstruction-based Auto-HAD model. Supporting libraries such as NumPy and SciPy are employed for numerical computations and matrix operations, while Scikit-learn is used for evaluation metrics, including ROC curve generation and AUC computation.

Model training and experimentation are conducted on GPU-enabled platforms such as Google Colab, Kaggle, or local systems equipped with CUDA-supported GPUs. These platforms provide the necessary computational power to process hyperspectral data efficiently and significantly reduce training time. Additionally, version control and project management are maintained using GitHub, ensuring proper documentation, collaboration, and reproducibility of results.

Source Collections

The proposed hyperspectral anomaly detection system is developed and evaluated using well-established benchmark datasets acquired from advanced airborne and satellite sensors such as AVIRIS (Airborne Visible/Infrared Imaging Spectrometer) and ROSIS (Reflective Optics System Imaging Spectrometer). These datasets are widely recognized in the remote sensing community for their high spectral resolution and reliability in evaluating anomaly detection algorithms. Each dataset consists of hyperspectral image cubes, where every pixel contains detailed spectral information across hundreds of contiguous bands, enabling precise

characterization of materials based on their spectral signatures. In addition to spectral data, corresponding ground truth maps are provided, which identify the location of anomalous targets within the scene, facilitating accurate performance evaluation [9].

The datasets used in this project include multiple real-world scenarios such as Los Angeles-1, Los Angeles-2, Gulfport, Texas Coast, Cat Island, and Pavia (The Source link for the dataset is https://github.com/jeline0110/GT-HAD/tree/main/data). These datasets vary significantly in terms of spatial dimensions, number of spectral bands, and the number of anomaly targets present. For instance, the Los Angeles datasets contain aircraft-based anomalies embedded within urban backgrounds, while the Gulfport and Texas Coast datasets represent coastal environments with complex spectral variability. The Cat Island dataset includes extremely sparse anomalies, making detection particularly challenging, and the Pavia dataset, captured using the ROSIS sensor, provides high-resolution urban imagery with subtle spectral variations [10]. This diversity ensures that the proposed models are tested under different environmental conditions, enhancing their robustness and generalization capability.

Another important characteristic of these datasets is the presence of sub-pixel anomalies, where the target occupies only a very small portion of the image. This makes anomaly detection a highly challenging task, as the spectral signatures of these targets are often mixed with the surrounding background [12]. The hyperspectral data is typically stored in the form of multi-dimensional cubes, where two dimensions represent spatial information and the third dimension represents spectral bands. These cubes are processed using sliding window techniques to extract local spatial-spectral patches, which serve as input to the deep learning models. The availability of accurate ground truth labels enables the computation of evaluation metrics such as ROC curves and AUC scores, ensuring a reliable comparison of model performance. Overall, the use of these benchmark datasets provides a strong foundation for validating the effectiveness of the proposed GT-HAD and Auto-HAD models in detecting rare and complex anomalies.

The analysis of software and hardware requirements, combined with insights from the literature survey, highlights the necessity of adopting advanced deep learning frameworks and efficient computational platforms for hyperspectral anomaly detection. The literature review clearly indicates that traditional statistical and reconstruction-based methods face limitations in handling high-dimensional spectral data and preserving anomaly information, particularly in complex

background environments.

From the system requirement perspective, the need for GPU-accelerated platforms, scalable frameworks like PyTorch, and robust dataset handling mechanisms is essential for successful implementation. The integration of transformer-based architectures with gating mechanisms, as explored in recent research, provides significant improvements in background suppression and anomaly detection accuracy. Therefore, the proposed system, supported by appropriate software and hardware infrastructure, aligns with both the findings of the literature survey and the defined system requirements. This ensures the development of a reliable, efficient, and high-performance hyperspectral anomaly detection framework capable of addressing real-world challenges in complex geospatial environments.

In addition to the diversity of environmental conditions represented across the selected hyperspectral image collections, a systematic preprocessing pipeline is employed to ensure that the input is suitable for deep learning-based anomaly detection. The raw hyperspectral cubes often contain variations in illumination, sensor noise, and intensity ranges across different scenes, which can negatively impact model learning if not properly handled. To address this, spectral normalization is performed to scale pixel values into a consistent range, thereby improving numerical stability during training. Furthermore, the high-dimensional hyperspectral cubes are processed using a sliding window mechanism to extract local spatial–spectral patches that preserve both contextual and spectral information. These patches serve as the fundamental input units for the proposed models, enabling them to capture fine-grained spatial structures as well as subtle spectral variations. This localized representation is particularly important for detecting small or sub-pixel anomalies, as it allows the models to focus on detailed patterns that may otherwise be lost in full-image processing. Additionally, careful handling of boundary regions and overlap between patches ensures that no critical information is discarded during extraction. Overall, this preprocessing stage plays a vital role in enhancing feature representation, improving convergence behavior, and ensuring that both GT-HAD and Auto-HAD models can effectively learn meaningful patterns from complex hyperspectral scenes. Furthermore, this framework is designed with a strong emphasis on consistency to ensure a reliable comparison between the proposed models. Moreover, by maintaining consistent experimental settings across all datasets, the results obtained reflect the true strengths and limitations of each architecture.

CHAPTER – 3

PROPOSED METHODOLOGY

This chapter provides a comprehensive technical roadmap of the methodologies implemented to achieve high-fidelity anomaly detection and airspace surveillance. The system is engineered to handle the high dimensionality and non-linear spectral variability of hyperspectral data through a dual-model paradigm. The sections below detail the end-to-end workflow, beginning with rigorous data preprocessing and spectral cube extraction, followed by the architectural design of the Gated Contextual Transformer (GT-HAD) and the Autonomous Autoencoder (Auto-HAD). Finally, the chapter explicates the optimization strategies, residual post-processing techniques, and the performance benchmarking protocols used to validate the system's effectiveness in national security scenarios.

3.1 DESIGN METHODOLOGY

3.1.1 Data Acquisition and Preprocessing

The foundation of the detection pipeline lies in the precise preparation of high-dimensional spectral data. The following steps ensure computational efficiency and spectral integrity, enabling the models to effectively learn meaningful spatial–spectral patterns for accurate anomaly detection.

- HSI Benchmark Integration: The system loads standardized hyperspectral datasets (e.g., AVIRIS Los Angeles, Gulfport, and ROSIS Pavia) in .mat format. Each scene contains hundreds of contiguous spectral bands along with binary ground-truth maps for validation.
- Multidimensional Normalization**:** To ensure stable gradient descent and prevent band-wise dominance, each spectral cube is normalized using a min-max scaling approach, bringing all radiance values into a consistent [0, 1] range.
- Local Context Extraction (Sliding Window): Instead of processing the entire wide-area image at once, a sliding window mechanism is utilized to partition the data into 3D HSI cubes (typically $9 \times 9 \times D$ or $15 \times 15 \times D$). This allows the model to process local spatial-spectral textures, which is essential for isolating aircraft that only occupy a few pixels in a massive scene.

3.1.2 Proposed Architectural Design

The project implements a "Competitive Model Strategy" by utilizing two distinct deep learning paradigms:

The Gated Contextual Transformer (GT-HAD)

This architecture is designed to decouple the reconstruction of background and anomalies through a dual-branch system:

- The Anomaly-Focused Branch (AFB): Utilizes a "patch-wise feature redefinition" mechanism. It attempts to reconstruct a patch using the weighted sum of its neighbors. Since anomalous jets exhibit low similarity to their surroundings, the AFB fails to reconstruct them correctly, thereby isolating the anomaly.
- The Background-Focused Branch (BFB): Leverages patch-level self-attention to reinforce the consistency of the terrestrial or atmospheric background.
- Intelligent Gating Unit (CMM): The Content-Matching Method (CMM) serves as the control logic. It routes pixels based on reconstruction fidelity; if a reconstructed cube matches the original input with high precision, the gate transitions it from AFB to BFB for further enhancement.

The Autonomous Reconstruction Network (Auto-HAD)

This model operates as a fully unsupervised, self-correcting autoencoder:

- Deep Image Prior (DIP): The model utilizes a skip-connected convolutional network that treats the architecture itself as a statistical regularizer for the HSI background.
- Adaptive Weighted Loss: The core innovation here is the periodic calculation of reconstruction residuals. Pixels with high residuals (potential anomalies) are assigned lower weights in the loss function, forcing the network to ignore them and become an expert purely in background reconstruction.

3.1.3 Training and Optimization Protocols

The training phase is designed to ensure the autonomous separation of spectral outliers without manual labeling:

- Optimizer Configuration: We employ the Adam optimizer with a tuned learning rate of 1×10^{-3} and a weight decay of 0 for precise reconstruction.
- Iteration Gating: Gating states and weight masks are updated periodically (e.g., every 25 or 100 iterations), ensuring that as the model moves toward convergence, it becomes increasingly better at suppressing anomalies.
- Spectral-Spatial Loss: The Mean Square Error (MSE) loss is calculated on the reconstructed cubes to minimize the distance between the original background and the model's output.

3.1.4 Anomaly Refinement and Post-Processing

Once the reconstruction is complete, the detection maps undergo refinement to improve clarity for human operators:

- Residual Computation: The pixel-wise anomaly score is calculated by measuring the squared difference between the original cube (x_i) and the reconstructed cube ($\hat{x}_i$).
- Residual Diffusion (RD): To mitigate the "blocking effect" and the mixing of pixels at target boundaries, a 3D average pooling filter ($3 \times 3 \times 5$ kernel) is applied. This diffuses high-intensity anomaly signals, making aircraft signatures sharper and the background flatter.
- Fold-and-Fusion: The final anomaly map (M) is generated using a fold operation that aggregates overlapping patch-level results via average fusion, resulting in a seamless, high-contrast spatial map.

3.1.5 Accuracy Assessment and Validation

To satisfy the rigorous standards of defense intelligence, the system's performance is quantitatively assessed using:

- ROC Curve Analysis: Plotting the Probability of Detection (PD) against the Probability of False Alarm (PF).
- AUC Score Calculation: Computing the Area Under the Curve to provide a single, robust diagnostic metric. A target score of > 0.98 is utilized as the benchmark.

3.2 SYSTEM ARCHITECTURE DIAGRAM (DESCRIPTION OF ALL THE MODULES)

This section describes the modular design of the two primary architectures implemented in this research. Each module serves a specific mathematical role in ensuring that the background is reconstructed with high fidelity while anomalous targets remain suppressed. The system architecture consists of hyperspectral anamoly detection as illustrated in Figure 3.1.

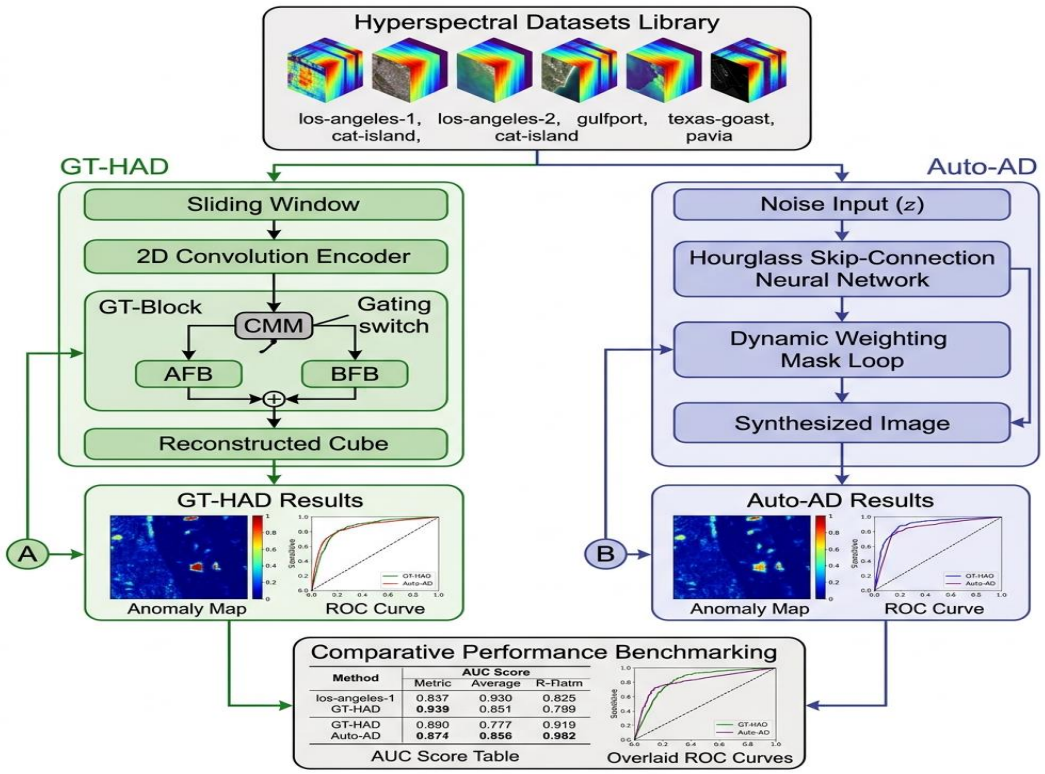

Figure 3.1: Architecture Diagram

3.2.1 Modules of the Gated Contextual Transformer (GT-HAD)

The GT-HAD system is composed of four principal functional modules that work in a closed loop during the training phase.

Spatial-Spectral Feature Encoding Module

The HSI cubes, after extraction, are first processed by this module. It utilizes a 3 × 3 convolutional layer to map the raw spectral radiance of the HSI cube (xi) into a high-dimensional latent feature map (fi ∈ ℝ^(H × W × C)). This module compresses the spectral dimension while preserving the spatial texture, creating an optimized representation for the subsequent Transformer blocks.

Gated Dual-Branch Network (GDBN)

This is the core decision-making unit of the architecture. It consists of two specialized pathways:

- The Anomaly-Focused Branch (AFB): This module implements a "feature redefinition" protocol at the patch level. It partitions the feature map into non-overlapping patches and calculates similarity weights between a central patch and its neighbors. By redefining the central patch as a weighted combination of its context, it effectively "erases" aircraft signatures that do not match the local background distribution.
- The Background-Focused Branch (BFB): This module utilizes a patch-level self-attention mechanism. It treats the patches as nodes in a graph and computes the spatial-spectral correlation across the entire cube. This reinforces the consistency of legitimate background features (such as soil, forest, or water), ensuring they are reconstructed with high precision.

Contextual Feature Matching Module (CFMM / CMM)

The CMM acts as the "intelligent gatekeeper." It records the gating state (gi) for every cube. During training, it computes the Euclidean distance between the reconstructed cube and the original dataset. If the model determines that a cube has been reconstructed with high fidelity (indicating it is a background component), the CMM transitions the gate to the BFB. If the reconstruction error is high (indicating an anomaly or content loss), the gate keeps the cube in the AFB.

Global Feature Refinement Module (FFN)

This Processed features from the GDBN are passed through a feed-forward network (FFN) consisting of layer normalization (LN) and a multi-layer perceptron (MLP). This module utilizes a GELU activation function and a channel-expansion strategy to capture the high-order non-linearities of the spectral data, followed by a skip connection to ensure gradient stability.

3.2.2 Modules of the Autonomous Reconstruction Network (Auto-HAD)

The Auto-HAD framework focuses on architectural priors and adaptive constraints to achieve unsupervised separation

Deep Image Prior (DIP) Backbone Module

This module utilizes a fully convolutional skip network (autoencoder). It consists of a contracting path (encoder) that reduces spatial dimensionality to capture global context and an expansive path (decoder) that restores the spectral radiance. The skip connections across the

architecture ensure that fine-grained spatial details are transferred directly from the input to the output, preventing the "blurring" of boundaries.

Adaptive Spatial Weighting Module

The "autonomous" logic is contained within this module. It periodically calculates a spatial importance mask (W) by evaluating the current reconstruction residuals. It identifies pixels with extremely high errors—which correspond to anomalous aircraft—and assigns them a near-zero weighting value. This effectively "hides" the anomalies from the network's optimization path, forcing the model to focus only on learning the background patterns.

Weighted Optimization Module

This module computes the final Weighted Mean Square Error (WMSE) loss. By multiplying the reconstruction error map by the adaptive spatial weights, it ensures that the backpropagation process only updates the network weights for background pixels. This module enables the system to operate autonomously without requiring any human-provided labels.

On applying these two models on the datasets we will get residual maps showing anamolies at the end of the architecture. Later we compare these models to declare which is the best one in performing anomalies detection in HSI data.

3.3 DESCRIPTION OF ALGORITHMS

This section presents a detailed description of the algorithms utilized in the proposed system, covering both conventional machine learning techniques and advanced deep learning approaches. The selected algorithms include sequence modeling methods such as Bi-LSTM, optimization techniques like FHO optimization, probabilistic modeling using CA-Markov, ensemble learning through Random Forest, and state-of-the-art hyperspectral anomaly detection models including GT-HAD and Auto-HAD. Each algorithm is chosen based on its ability to address specific challenges such as temporal dependency modeling, feature optimization, spatial prediction, classification, and anomaly detection in high-dimensional data. The algorithms are explained with their respective mathematical formulations and step-by-step procedures. While traditional models contribute to baseline analysis and predictive capabilities, the deep learning-based GT-HAD and Auto-HAD models.

3.3.1 Bi-LSTM Algorithm

The detailed Bi-LSTM algorithm steps containing equations are shown in below Table 3.1.

Table 3.1 Bi-LSTM Algorithm

Input:

- X: A sequence of 12 features (e.g., 6 vegetation indices like NDVI, 6 spectral bands) for a land pixel, shaped as (1, 12).
- K: Number of classes (6: Urban, Agricultural Land, Forest, Water Bodies, Barren Land, Grassland).

Output:

- Predicted probabilities $P(j \mid x)$for each class $j \in \{1, \dots, K\}$for the test instance x.

Method:

1. Initialize Bidirectional LSTM Layer with 128 Units:

- For the sequence $X = [x_1, x_2, \dots, x_{12}]$:

a. Forward LSTM Pass: Process $X$from $t = 1$to $t = 12$

- For each time step t:

Compute forget gate:

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}^{fwd}, x_t\right] + b_f\right) \quad \text{[Eq-3.1]}$$

Compute input gate:

$$i_t = \sigma\left(W_i \cdot \left[h_{t-1}^{fwd}, x_t\right] + b_i\right) \quad \text{[Eq-3.2]}$$

Compute new memory:

$$\tilde{C}_t = \tanh\left(W_C \cdot \left[h_{t-1}^{fwd}, x_t\right] + b_C\right) \quad \text{[Eq-3.3]}$$

Update memory:

$$C_t^{fwd} = f_t \cdot C_{t-1}^{fwd} + i_t \cdot \tilde{C}_t \quad \text{[Eq-3.4]}$$

Compute output gate:

$$o_t = \sigma\left(W_o \cdot \left[h_{t-1}^{fwd}, x_t\right] + b_o\right) \quad \text{[Eq-3.5]}$$

Output:

$$h_t^{fwd} = o_t \cdot \tanh\left(C_t^{fwd}\right) \quad \text{[Eq-3.6]}$$

b. Backward LSTM Pass: Process $X$from $t = 12$to $t = 1$

- Apply the same LSTM equations as above (Equations 3.1 to 3.6), but in reverse, yielding h_t^{bwd}.

c. Concatenate Outputs:

$$h = \left[h_{12}^{fwd}, h_1^{bwd}\right] \quad \text{[Eq-3.7]}$$

2. Apply Dropout (First Layer):

- Randomly set 30% of the elements in $h$to 0 (Dropout rate = 0.3).

3. Dense Layer with 64 Units (ReLU Activation):

Compute:

$$z = W_d \cdot h + b_d \quad \text{[Eq-3.8]}$$

Apply ReLU:

$$h_{dense} = \max(0, z) \quad \text{[Eq-3.9]}$$

4. Apply Dropout (Second Layer):

- Randomly set 30% of the elements in $h_{dense}$to 0 (Dropout rate = 0.3).

5. Final Dense Layer with Softmax:

Compute scores for each class:

$$z_{final} = W_{final} \cdot h_{dense} + b_{final} \quad \text{[Eq-3.10]}$$

Apply Softmax to get probabilities:

$$P(j \mid x) = \frac{e^{z_{final,j}}}{\sum_{k=1}^{K} e^{z_{final,k}}}, \text{for } j = 1, \dots, K \quad \text{[Eq-3.11]}$$

6. Return Probabilities:

- Output the probability vector

$$[\boldsymbol{P}(\mathbf{1} \mid \boldsymbol{x}), \boldsymbol{P}(\mathbf{2} \mid \boldsymbol{x}), \dots, \boldsymbol{P}(\boldsymbol{K} \mid \boldsymbol{x})]$$

3.3.2 FHO Algorithm

The detailed FHO algorithm steps containing equations are shown in below Table 3.2. The FHO (Flamingo-Hyena Optimization technique) algorithm is employed to optimize model parameters by efficiently exploring this solution space . The optimization process enhances the overall performance and accuracy of the model.

Table 3.2 FHO Algorithm

INPUT:

- w: Current model weights (settings).
- g: Gradients (how wrong the model is).
- lr: Learning rate (0.001).
- β_1: First moment decay (0.9).
- β_2: Second moment decay (0.999).
- ϵ: Small value to avoid division by zero (1×10^{-7}).

OUTPUT:

- Updated weights $w$for better predictions.

Method:

1. Initialize Adam Parameters:

- Set $m_0 = 0$(first moment, average of gradients).
- Set $v_0 = 0$(second moment, average of squared gradients).
- Set $t = 0$(time step).
- Set best_weights = NULL, best_loss = ∞.

2. Perform Adam Update:

- Increment time step:

$$t = t + 1$$

Update first moment:

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g \qquad \text{[Eq-3.12]}$$

Update second moment:

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g^2 \qquad \text{[Eq-3.13]}$$

Correct bias in moments:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \qquad \text{[Eq-3.14]}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \qquad \text{[Eq-3.15]}$$

Update weights:

$$w = w - lr \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \qquad \text{[Eq-3.16]}$$

3. Apply Flamingo Search (FSA-Inspired Exploration):

- If random number < 0.15:

Generate perturbation:

$$pert = random(-lr \cdot 0.1,\ lr \cdot 0.1) \qquad \text{[Eq-3.17]}$$

Update weights:

$$w = w + pert \cdot w \qquad \text{[Eq-3.18]}$$

4. Apply Spotted Hyena (SHO-Inspired Exploitation):

- If best_weights ≠ NULL and random number < 0.1:

Compute direction:

$$dir = best_{weights} - w \qquad \text{[Eq-3.19]}$$

Update weights:

$$w = w + 0.05 \cdot dir \qquad \text{[Eq-3.20]}$$

5. Update Best Weights if Loss Improves:

- Compute current loss L(e.g., sparse categorical cross-entropy).
- If $L <$ best_loss:
  - Set best_loss = L
  - Set best_weights = w

6. Return Updated Weights:

- Output the updated w.

3.3.3 CA_Markov_Hybrid_Prediction

The detailed CA-Markov algorithm (that is used for future prediction) steps containing equations are shown in below Table 3.3. This algorithm integrates Cellular Automata with Markov chain modeling to capture both spatial dynamics and temporal transition probabilities for accurate future state prediction. By combining neighborhood influence with probabilistic state transitions, it effectively models complex spatial changes over time. It is particularly useful in scenarios such as land-use and land-cover prediction, where both spatial dependency and temporal evolution play a crucial role. This makes it a reliable approach for simulating realistic future scenarios in dynamic geospatial environments.

Table 3.3: CA-Markov Algorithm

INPUT:

- $C_{2020}(x)$: Classified land-use image for the year 2020.
- $C_{2025}(x)$: Classified land-use image for the year 2025.
- Ω: Study area boundary used as mask for regional statistics.
- K: Total number of land-use classes:

  $K = \{\text{Built-up, Cropland, Forest, Grassland, Shrubland, Barren Land, Water, Wetland}\}$

- $a_{px} = 0.0009\ \text{km}^2$: Pixel area.
- K_{CA}: Square neighborhood kernel of radius 2 pixels.

OUTPUT:

- Predicted land-use map for 2030:

$$C_{2030}(x)$$

METHOD:

Step 1 — Compute Transition Areas Between 2020 and 2025

For each class pair (i, j), where $i, j \in \{1, \dots, K\}$:

Create transition mask:

$$M_{ij}(x) = [C_{2020}(x) = i] \wedge [C_{2025}(x) = j] \qquad \text{[Eq-3.21]}$$

Compute transition area:

$$A_{ij} = \sum_{x \in \Omega} M_{ij}(x) \cdot a_{px} \qquad \text{[Eq-3.22]}$$

This results in a $K \times K$ transition matrix representing land-cover changes.

Step 2 — Derive Markov Transition Probabilities

For each original class i:

Compute total outgoing area:

$$S_i = \sum_{j=1}^{K} A_{ij} \qquad \text{[Eq-3.23]}$$

Compute transition probability:

$$P_{ij} = \begin{cases} \dfrac{A_{ij}}{S_i}, & S_i > 0 \\ 0, & S_i = 0 \end{cases} \qquad \text{[Eq-3.24]}$$

The transition matrix $P$is stochastic, satisfying:

$$\sum_{j=1}^{K} P_{ij} = 1 \qquad \text{[Eq-3.25]}$$

Step 3 — Compute Class-wise Areas for 2025

For each class i:

$$a_i^{(2025)} = \sum_{x \in \Omega} [\, C_{2025}(x) = i] \cdot a_{px} \qquad \text{[Eq-3.26]}$$

The area vector is represented as:

$$a^{(2025)} = \left\{ a_1^{(2025)}, \dots, a_K^{(2025)} \right\} \qquad \text{[Eq-3.27]}$$

Step 4 — Predict Class-wise Areas for 2030

The projected land-use distribution is computed as:

$$a^{(2030)} = a^{(2025)} \cdot P \qquad \text{[Eq-3.28]}$$

Step 5 — Allocate Predicted Areas Using Cellular Automata

For each transition $i \to j$, where $i \neq j$:

Compute expected transition area:

$$E_{ij} = a_i^{(2025)} \cdot P_{ij} \qquad \text{[Eq-3.29]}$$

Compute required pixels:

$$N_{\ ij} = \text{round}\left(\frac{E_{ij}}{a_{px}} \right) \qquad \text{[Eq-3.30]}$$

Neighborhood Suitability Computation

Compute neighborhood dominance:

$$S_j(x) = \sum_{y \in K_{CA}(x)} [\, C_{2025}(y) = j] \qquad \text{[Eq-3.31]}$$

Compute suitability:

$$\text{Suit}_{i \to j}(x) = S_j(x) \cdot [C_{2025}(x) = i] \qquad \text{[Eq-3.32]}$$

N_{ij}pixels are reassigned:

$$C_{2030}(x) = j \qquad \text{[Eq-3.33]}$$

Final Output:

The final simulated land-cover map for the year 2030 is:

$$C_{2030}(x)$$

3.3.4. Algorithm: Random Forest Classification

The random Forest algorithm steps containing equations are shown in below Table 3.4

Table 3.4: Random Forest Algorithm

Input

Training dataset T_{train}containing feature vector X

- Model settings taken from the code:
  - Number of trees T
  - bagFraction
  - variablesPerSplit (m)
  - minLeafPopulation

Output

- Final predicted class $h(X)$for a pixel (or sample) X,
  or class probability vector $P(c \mid X)$over all classes.

Method

For each classifier with T trees:

1. For tree $t = 1 \ldots T$:

Draw bootstrap sample $B_t$from T_{train}(size determined by **bagFraction**).

Grow tree $h_t$by recursively splitting nodes:

- At each node, select $m$features uniformly at random (**variablesPerSplit**).
- For candidate splits, compute impurity (Gini):

$$G = 1 - \sum_{c=1}^{K} p_c^2 \qquad \text{[Eq-3.34]}$$

$$p_c = \frac{\#\text{class } c}{n} \qquad \text{[Eq-3.35]}$$

For split into left (L) and right (R):

$$G_{split} = \frac{n_L}{n} G_L + \frac{n_R}{n} G_R \qquad \text{[Eq-3.36]}$$

$$\Delta G = G - G_{split} \qquad \text{[Eq-3.37]}$$

Choose the split that maximizes ΔG.

Stop when **minLeafPopulation is reached** or other stopping criteria are satisfied.

Leaf nodes store empirical class distribution:

$$P(c \mid X \in \text{leaf})$$

2. Tree Output for Pixel X:

Predicted class $h_t(X)$or probability vector $P_t(c \mid X)$

3.3.5 GT-HAD Algorithm

The GT-HAD algorithm steps containing equations are shown in below Table 3.5

Table 3.5 GT-HAD Algorithm

Input

$$X \in \mathbb{R}^{H \times W \times D}, X = \{x_i\}_{i=1}^{N}$$

- X: Hyperspectral image
- x_i: Extracted overlapping HSI cubes

Initialization

$$G = [g_1, g_2, \ldots, g_N] = 0$$

- Initialize gating states; all features start in anomaly-focused processing.

Output

- M: Anamoly Map

Training Stage

Step 1: Sliding Window Extraction

HSI is partitioned into overlapping cubes using a fixed window and stride.

$$x_i = \text{SlidingWindow}(X) \qquad \text{[Eq-3.38]}$$

Step 2: Feature Encoding

Each cube is mapped into a spatial–spectral feature space.

$$f_i = Conv_{3\times3}(x_i) \qquad \text{[Eq-3.39]}$$

- f_i: Feature representation of cube x_i

Step 3: Gating Decision

Feature maps are dynamically routed based on gating state.

$$\widetilde{f_i} = \begin{cases} AFB(LN(f_i)), & g_i = 0 \\ BFB(LN(f_i)), & g_i = 1 \end{cases} \quad \text{[Eq-3.40]}$$

- AFB: Suppresses anomaly features
- BFB: Enhances background features

Step 4: Patch Partitioning

Feature map is divided into smaller patches for local similarity modeling.

$$f_i \rightarrow \{p_a\}_{a=1}^{J_A} \quad \text{[Eq-3.41]}$$

- p_a: Patch representation

Step 5: Similarity Computation

Similarity between patches is computed using learned projections.

$$S(p_a, p_j) = \langle p_a W_A, p_j W_A \rangle \quad \text{[Eq-3.42]}$$

- Measures content similarity between neighboring patches

Step 6: Weight Calculation

Softmax is applied to normalize similarity scores.

$$\omega_{aj} = \frac{\exp(S(p_a, p_j))}{\sum_{z \in Z} \exp(S(p_a, p_z))} \quad \text{[Eq-3.43]}$$

- ω_{aj}: Contribution weight of neighboring patch

Step 7: Patch Redefinition

Each patch is reconstructed using weighted neighboring patches.

$$\widehat{p_a} = \sum_{j \in Z} \omega_{aj}\, p_j \quad \text{[Eq-3.44]}$$

- Removes anomaly information due to low similarity

Step 8: Patch Reconstruction

All updated patches are combined to reform the feature map.

$$\widetilde{f_i} = I\left(\{\widehat{p_a}\}_{a=1}^{J_A}\right) \quad \text{[Eq-3.45]}$$

- Restores full spatial structure from patches

Step 9: Background Feature Tensor

Feature map is reshaped for attention-based processing.

$$T_B \in \mathbb{R}^{J_B \times L_B \times C} \quad \text{[Eq-3.46]}$$

- T_B: Tensor representation of patches

Step 10: Self-Attention

Background features are enhanced using attention mechanism.

$$Q = T_B W_B, K = T_B W_B, V = T_B \quad \text{[Eq-3.47]}$$

$$\widetilde{f_\iota} = Softmax(QK^T)V \quad \text{[Eq-3.48]}$$

Step 11: Feed Forward Network

Non-linear transformation with residual connection is applied.

$$\widehat{f_\iota} = MLP(LN(\widetilde{f_\iota})) + \widetilde{f_\iota} \quad \text{[Eq-3.49]}$$

Step 12: Reconstruction

Reconstructed cube is generated from refined features.

$$\widehat{x_\iota} = Conv_{3\times3}(\widehat{f_\iota}) \quad \text{[Eq-3.50]}$$

- $\widehat{x_\iota}$: Reconstructed HSI cube

Step 13: Loss Function

Model is trained using reconstruction error minimization.

$$L(\theta) = \frac{1}{B}\sum_{i=1}^{B} \| x_i - \widehat{x_\iota} \|^2 \quad \text{[Eq-3.51]}$$

Step 14: Content Matching

Reconstruction similarity is used to update gating states.

$$x_v = \arg\min \quad || \widehat{x_\iota} - x_j || \quad \text{[Eq-3.52]}$$

$$g_i = \begin{cases} 1, & v = i \\ 0, & v \neq i \end{cases} \quad \text{[Eq-3.53]}$$

Step 15: Backpropagation

Network parameters and gating states are updated iteratively until convergence.

Detection Stage

Step 16: Residual Computation

Difference between input and reconstructed cube is computed.

$$r_i = (x_i - \widehat{x_\iota})^2 \quad \text{[Eq-3.54]}$$

Step 17: Residual Map

Residual values are aggregated across spectral bands.

$$m_i = \sum_{bands} r_i \qquad \text{[Eq-3.55]}$$

- Produces pixel-wise anomaly response

Step 18: Aggregation

All residual maps are combined into final anomaly map.

$$M = F(\{m_i\}) \qquad \text{[Eq-3.56]}$$

- Combines overlapping regions

Step 19: Residual Diffusion

Spatial smoothing enhances anomaly regions.

$$e_i = AvgPool3D(r_i) \qquad \text{[Eq-3.57]}$$

Output

$$M$$

- Final anomaly map highlighting abnormal regions in the HSI.

3.3.6 Auto-HAD Algorithm

The Auto-HAD algorithm steps containing equations are shown in below Table 3.6

Table 3.6 Auto-HAD Algorithm

Input

Hyperspectral Image:

$$X \in \mathbb{R}^{H \times W \times D}$$

Network Input (Noise Tensor):

$$Z \in \mathbb{R}^{H \times W \times D}$$

Initialization

Weight Mask:

$$W \in \mathbb{R}^{H \times W}$$

initialized to 1 (equal importance for all pixels).

Parameters: Network weights θ for the Fully Convolutional Skip-Net.

TRAINING STAGE (Self-Supervised)

Step 1 — Noise Propagation

Generate the reconstruction from the noise input Zthrough the Autoencoder:

$$\hat{X} = f_\theta(Z) \qquad \text{[Eq-3.58]}$$

Step 2 — Feature Extraction & Skip-Connections

The encoder extracts hierarchical spatial-spectral features, while skip-connections preserve fine-grained structural details:

$$\hat{X} = \text{SkipNet}(Z, \theta) \qquad \text{[Eq-3.59]}$$

Step 3 — Adaptive Weighting (Adaptive Weighted Loss)

Every $k$iterations (e.g., 100), update the importance of each pixel:

Pixel-wise Residual:

$$R(x, y) = \mid X(x, y) - \hat{X}(x, y) \mid_2^2 \qquad \text{[Eq-3.60]}$$

Foreground Suppressing Weight:

$$W'(x, y) = \max(R) - R(x, y) \qquad \text{[Eq-3.61]}$$

Normalization:

$$W(x, y) = \frac{W' - \min(W')}{\max(W') - \min(W')} \qquad \text{[Eq-3.62]}$$

This assigns high weights to background (low error) and low weights to anomalies (high error).

Step 4 — Weighted Loss Function

Optimize the network to reconstruct the background specifically by masking out anomalous regions:

$$\mathcal{L}(\theta) = \frac{1}{B}\sum_{i=1}^{B} \mid (\hat{X}_i - X_i) \odot W \mid_2^2 \qquad \text{[Eq-3.63]}$$

(where $\odot$is element-wise multiplication)

Step 5 — Backpropagation

Update parameters θusing the Adam optimizer to minimize $\mathcal{L}(\theta)$.

DETECTION STAGE

Step 6 — Final Residual Generation

Once training converges, compute the final difference between the original and reconstructed images:

$$\epsilon(x, y) = \mid X(x, y) - \hat{X}(x, y) \mid_2^2 \qquad \text{[Eq-3.64]}$$

Step 7 — Anomaly Scoring

Sum the squared residuals across all spectral bands to get the spatial anomaly map.

FINAL OUTPUT

Anomaly Map $\boldsymbol{M}$

3.4 DESCRIPTION OF DATASET AND TOOLS

3.4.1 DATASET

The models are trained and validated using benchmark hyperspectral datasets obtained from AVIRIS and ROSIS sensors. These datasets consist of high-dimensional spectral cubes along with corresponding ground truth anomaly maps, which serve as a reference for evaluating model performance. The study includes multiple datasets such as Los Angeles-1, Los Angeles-2, Gulfport, Texas Coast, Cat Island, and Pavia, each representing different environmental conditions including urban, coastal, and island regions.

Each dataset contains hundreds of spectral bands, allowing detailed analysis of spectral signatures, while the spatial dimensions vary across datasets to ensure robustness of the models. The anomalies present in these datasets are typically small, rare targets such as aircraft or unusual objects, making detection a challenging task. The use of such diverse and complex datasets ensures that the proposed models are evaluated under realistic conditions and can generalize effectively across different scenarios. The Summary of dataset is shown bwlow in Table 3.7. The Source link for the dataset is https://github.com/jeline0110/GT-HAD/tree/main/data .

Table 3.7: Dataset Summary

DATASET	BANDS	DIMENSIONS	TARGET
los-angeles-1	**100**	**100x205**	**144**
los-angeles-2	**100**	**100x205**	**87**
Gulfport	**100**	**100x191**	**60**
texas-goast	**100**	**100x204**	**67**
cat-island	**150**	**150x188**	**19**
pavia	**150**	**150x102**	**68**

The images of different datasets in RGB view is shown below in Figure 3.2

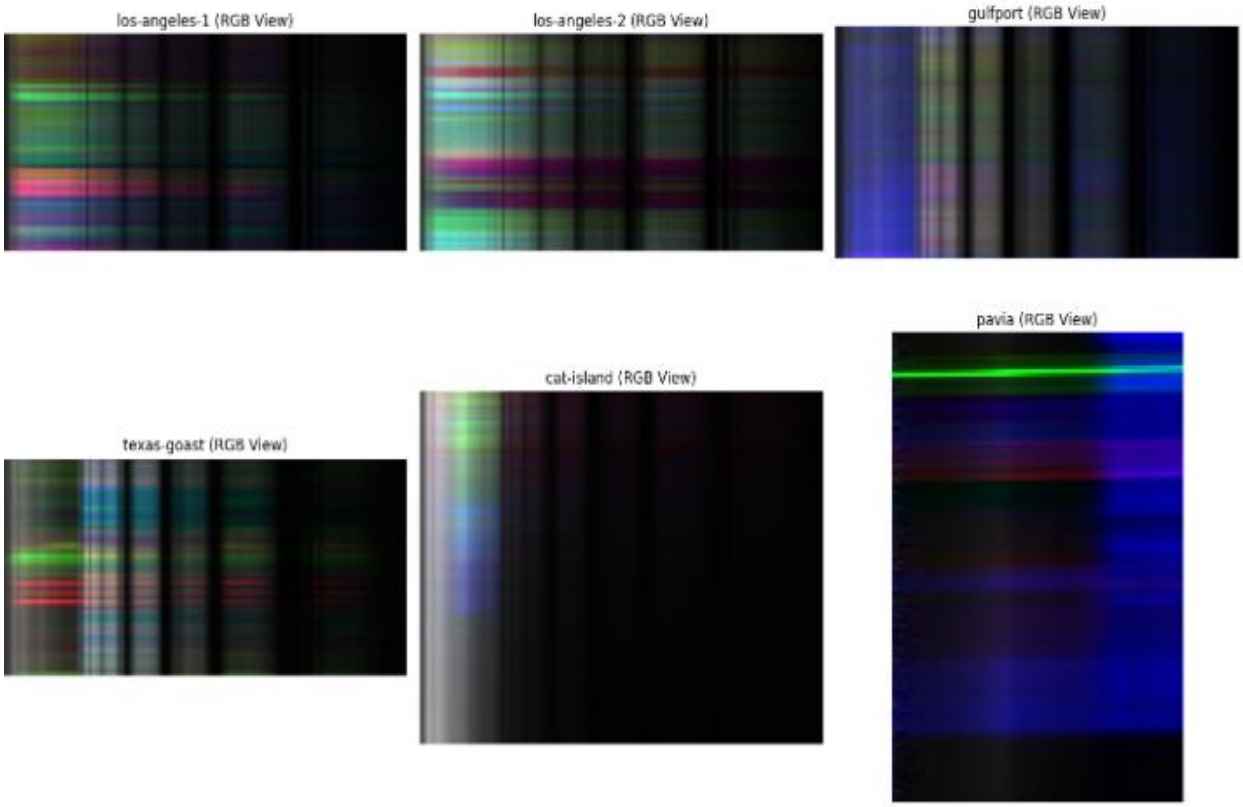

Figure 3.2 : RGB View of Dataset

3.4.2 TOOLS

Python

Python is used as the primary programming language for implementing the hyperspectral anomaly detection system. It provides flexibility, ease of development, and extensive support for scientific computing and deep learning. Its rich ecosystem of libraries enables efficient handling of high-dimensional hyperspectral data and seamless integration with machine learning frameworks.

PyTorch

PyTorch is the core deep learning framework used for building and training the GT-HAD and Auto-HAD models. It supports dynamic computation graphs and GPU acceleration, making it suitable for implementing complex architectures such as transformer-based models and convolutional autoencoders. PyTorch also allows efficient backpropagation and optimization

during training.

NumPy

NumPy is used for numerical computations and efficient manipulation of multidimensional arrays. It plays a crucial role in handling hyperspectral data cubes, performing matrix operations, and supporting preprocessing tasks such as normalization and data transformation.

SciPy

SciPy is utilized for advanced scientific computations and processing tasks. It supports operations related to signal processing, data handling, and mathematical functions required during preprocessing and analysis of hyperspectral datasets.

Scikit-learn

Scikit-learn is used for performance evaluation and validation of the models. It provides tools for generating ROC curves, computing AUC scores, and analyzing the effectiveness of anomaly detection methods.

Kaggle Platform

Kaggle is used as the primary platform for model training and experimentation. It provides a cloud-based environment with GPU support, enabling efficient execution of computationally intensive deep learning models. It also facilitates dataset management, code execution, and result visualization within an integrated notebook environment.

GitHub

GitHub is used for version control and project management. It helps in maintaining code consistency, tracking changes, and enabling collaboration. It also ensures reproducibility of results by preserving the complete implementation workflow.

In summary, the methodology outlined in this chapter provides a rigorous and scalable framework for the identification of high-altitude spectral anomalies. By integrating the gated decision-making capabilities of the Gated Transformer with the unsupervised autonomy of the Autonomous Autoencoder, the system effectively overcomes the traditional limitations of background modeling in high-dimensional imagery. These synergistic deep-reconstruction paradigms, supported by adaptive weighting and residual refinement protocols.

CHAPTER – 4

RESULTS AND OBSERVATIONS

The results section presents the comprehensive outcomes obtained from the progressive development of the project, starting from this project, which focused on deep learning-based LULC classification using a BiLSTM model, later we introduced CA-Markov modeling for predictive land cover analysis, and finally extending to the project, which implements advanced hyperspectral anomaly detection using deep learning architectures such as GT-HAD and Auto-HAD. The comparative analysis further highlights the superiority and robustness of the proposed models in terms of accuracy, prediction capability, and anomaly detection performance across diverse scenarios.

4.1 DESCRIPTION OF RESULTS

4.1.1 Bi-LSTM Model and FHO Optimization

The results of the land cover classification project for Vijayawada, spanning 2010 to 2020, are presented through various visualizations and quantitative measures, reflecting the effectiveness of the FHO-BiLSTM model. The classified Land Use Land Cover (LULC) maps for 2010 and 2020 illustrate the spatial distribution of six classes: Urban, Agricultural Land, Forest, Water Bodies, Barren Land, and Grassland, with colors representing each category—red for Urban, green for Agricultural Land, purple for Barren Land, light green for Grassland, brown for Forest, and blue for Water Bodies. The 2010 map reveals a diverse landscape dominated by Agricultural Land and Barren Land, while the 2020 map shows a noticeable increase in Urban areas, indicating significant land use changes. The Change Map (2010-2020) highlights these shifts with black and white patterns, emphasizing regions of alteration. Training performance is depicted through accuracy and loss plots, where accuracy rises from 0.91 to 0.96 (training) and peaks at 0.9762 (validation), stabilizing at 0.9722, while loss decreases from 0.2356 to 0.1013 (training) and 0.1131 to 0.0632 (validation) over 10 epochs. Evaluation metrics report an accuracy of 0.9722, precision of 0.9725, recall of 0.9722, and F1-score of 0.9719, underscoring the model’s robustness. The land cover transition table quantifies changes, noting stable Urban areas at 5.50 sq.km, with notable shifts like Agricultural Land to Urban (6.48 sq.km) and Grassland to Urban (4.09 sq.km), though Water Bodies remain minimally represented. These results clearly indicate rapid urban expansion and a corresponding reduction in natural and semi-natural land.

Land Cover Maps (2010 and 2020)

The LULC maps for 2010 and 2020 provide a visual representation of land cover distribution across Vijayawada. The 2010 map showcases a mosaic of colors, with Agricultural Land (green) and Barren Land (purple) dominating, alongside sparse blue patches indicating Water Bodies. By 2020, the map reveals a marked expansion of Urban areas (red), accompanied by persistent Barren Land and reduced Water Bodies, reflecting urban growth and potential misclassification challenges. The following Figure 4.1 shows the LULC maps from the FHO-BiLSTM model, depicting six land cover classes (Urban, Agricultural Land, Forest, Water Bodies, Barren Land, Grassland) in Vijayawada from 2010 to 2020, highlighting accurate classification.

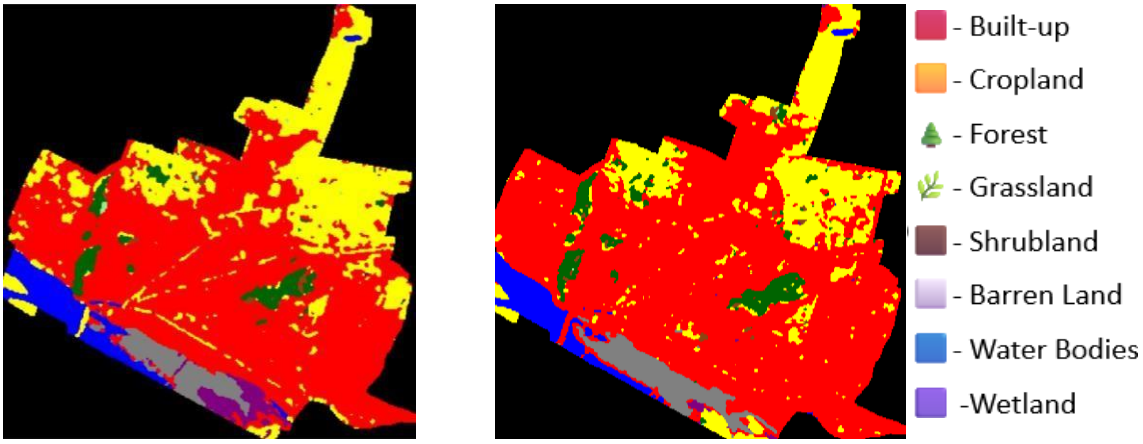

Figure 4.1 LULC Maps(2010 and 2020)

Accuracy and Loss Plots

The accuracy and loss plots track the model's training progress over 10 epochs. Accuracy plots show training accuracy climbing from 0.91 to 0.96 and validation accuracy peaking at 0.9762 before stabilizing at 0.9722. Loss plots indicate a decline from 0.2356 to 0.1013 (training) and 0.1131 to 0.0632 (validation), demonstrating consistent model convergence and reliability. The following Figure 4.2 shows the accuracy and loss plots for the FHO-BiLSTM model, demonstrating strong training performance with high accuracy (0.9762) over 10 epochs. The close alignment between training and validation curves indicates minimal overfitting and good generalization capability of the model. The steady reduction in loss further confirms that the model is effectively learning meaningful patterns from the input sequences. Additionally, the early stabilization of validation accuracy suggests that the model reaches optimal performance.

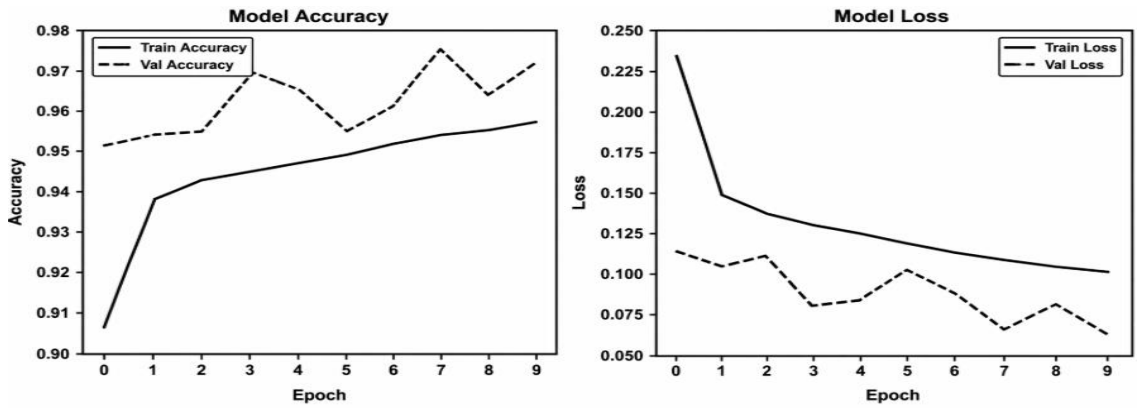

Figure 4.2 Training and Validation Accuracy & Training and Validation Loss

Evaluation Metrics

Evaluation metrics provide a quantitative assessment of the model's performance, with an accuracy of 0.9722, precision of 0.9725, recall of 0.9722, and F1-score of 0.9719. These values, derived from the validation set, affirm the model's high capability in classifying the six land cover classes. The following Table 4.1 shows the FHO-BiLSTM model's evaluation metrics, with accuracy (0.9722), precision (0.9741), recall (0.9722), and F1-score (0.9727), reflecting reliable classification.

Table 4.1 Bi-LSTM-FHO Evaluation Metrics

Metric	Value
Accuracy	0.9722
Precision	0.9725
Recall	0.9722
F1-Score	0.9719

Land Cover Transition Table

The transition table details land use changes from 2010 to 2020 in square kilometers. Notable figures include 5.50 sq.km of stable Urban areas, 6.48 sq.km of Agricultural Land converting to Urban, and 4.09 sq.km of Grassland shifting to Urban, alongside minimal Water Bodies representation, highlighting significant urban expansion and potential under-detection of aquatic areas. The following table 4.2 shows the land cover transition table, capturing shifts like Agricultural Land to Urban in Vijayawada from 2010 to 2020, as predicted by the FHO-BiLSTM

model. These transitions indicate a clear trend of land transformation driven by urban growth and changing land use patterns.

Table-4.2 Land Cover Transition table

From Class	**To Class**	**Area (Sq.Km)**
Urban	Urban	5.50
Urban	Barren Land	2.12
Urban	Grassland	3.46
Urban	Agricultural Land	6.70
Barren Land	Urban	1.33
Barren Land	Barren Land	3.26
Barren Land	Grassland	0.53
Barren Land	Agricultural Land	3.86
Grassland	Urban	4.09
Grassland	Barren Land	1.71
Grassland	Grassland	2.99
Grassland	Agricultural Land	6.49
Agricultural Land	Urban	6.48
Agricultural Land	Barren Land	3.24
Agricultural Land	Grassland	5.72
Agricultural Land	Agricultural Land	4.08

4.1.2 Random Forest and CA-Markov Model

The overall results of the project demonstrate successful historical land use and land cover (LULC) analysis and reliable future prediction for Vijayawada using a CA-Markov hybrid model integrated with Google Earth Engine and ArcGIS. Historical classification of multi-temporal satellite imagery (Landsat-5 for 2010, Landsat-8 for 2015, and Sentinel-2 for 2020 and 2025) achieved

overall accuracies ranging from 85.56% (2010) to a peak of 90% (2020), with a slight dip to 87.22% in 2025 attributed to seasonal variations and limited recent data availability. The classified maps reveal fluctuating built-up areas (44.656 km² in 2010 → 38.704 km² in 2025), significant cropland expansion especially in 2025 (14.349 km²), and minor declines in forest cover, reflecting rapid urbanisation followed by temporary agricultural intensification. The CA-Markov model, calibrated using transition probabilities derived from 2010–2025 intervals and validated with a Kappa coefficient of 0.92, accurately forecasted LULC for 2030, projecting a renewed increase in built-up area to 41.357 km², moderation of cropland to 11.031 km², and slight recovery in wetland and water bodies. These outcomes provide clear, visually interpretable LULC maps and quantitative area statistics that effectively highlight urban sprawl trends, enabling planners and policymakers to anticipate future land cover dynamics and formulate sustainable urban development strategies for Vijayawada.

Classified LULC Map of Vijayawada – 2010

The classified LULC map for 2010 in Figure 4.3, derived from Landsat-5 imagery using RandomForest in Google Earth Engine, shows Built-up areas (red) as the dominant class, forming the compact urban core of Vijayawada. Cropland (yellow) occupies extensive peripheral zones, while Forest (dark green) and Water Bodies (blue) are clearly visible along the Krishna River. Smaller patches of Barren Land (grey), Wetland (purple), and negligible Grassland/Shrubland complete the distribution, achieving an overall accuracy of 85.56%. This classification achieved an overall accuracy of 85.56%, with a kappa coefficient of 0.82, validated through stratified random sampling and confusion matrix analysis. The Built-up class covered approximately 28% of the total area. The LULC for the year 2010 is shown below in Figure 4.3.

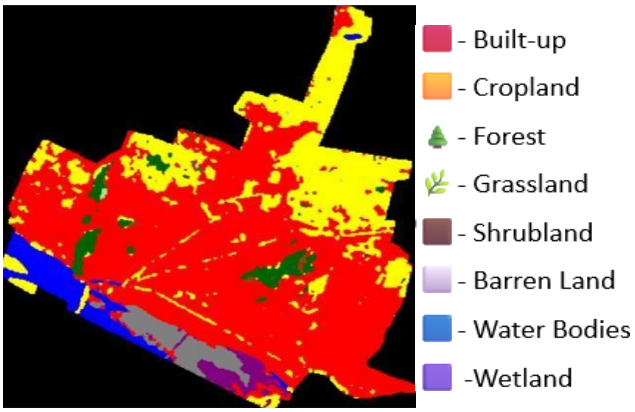

Figure 4.3 Classified LULC Map of Vijayawada – 2010

Classified LULC Map of Vijayawada – 2015

The classified LULC map for 2015 in Figure 4.4, generated from Landsat-8 imagery using Random Forest, displays a prominent expansion of Built-up areas (red) compared to 2010, indi. Cropland (yellow) remains extensive in peri-urban zones, while Forest (dark green) patches. Water Bodies (blue) along the Krishna River are clearly visible, with Barren Land (grey) and Wetland (purple) forming smaller clusters. The LULC map for the year 2015 is shown below in Figure 4.4

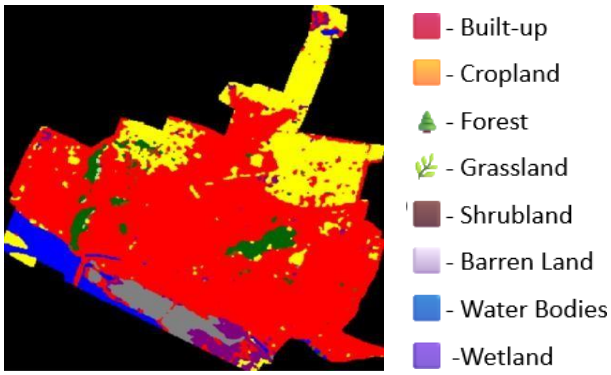

Figure 4.4 Classified LULC Map of Vijayawada – 2015

Classified LULC Map of Vijayawada – 2020

The classified LULC map for 2020 in Figure 4.5, derived from Sentinel-2 imagery using Random Forest in Google Earth Engine, exhibits a clear dominance of Built-up areas (red) with significant expansion over previous years reflecting intensified urbanization. Cropland (yellow) remains prominent in surroundings, The map achieved the highest overall accuracy of 90%, and the LULC for the year 2020 is shown below in Figure 4.5

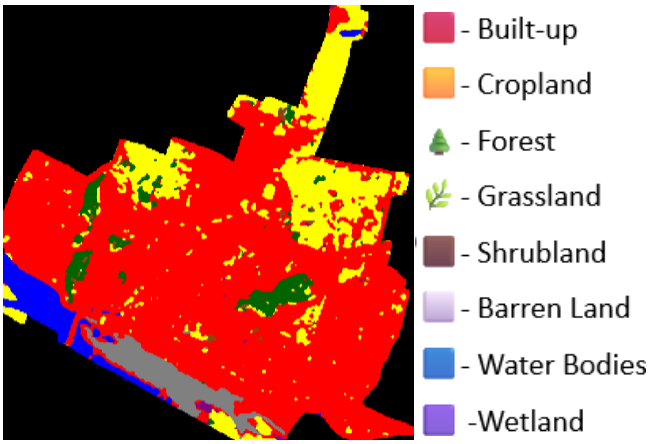

Figure 4.5 Classified LULC Map of Vijayawada – 2020

Classified LULC Map of Vijayawada – 2025

The LULC map for 2025 in Figure 4.6 reveals notable shifts driven by recent seasonal and developmental influences. Urban areas (red) show a temporary decline compared to 2020, contracting to approximately 38.7 km², possibly due to cloud-shadow effects or reclassification of peri-urban zones in the Sentinel-2 composite. Cropland (yellow) exhibits a significant surge to 14.349 km², indicating agricultural intensification, while wetlands (purple) expand markedly to 1.289 km², reflecting increased inundation along the Krishna River tributaries and improved detection of seasonal waterlogged areas. Water bodies (blue) and forest cover (dark green) remain relatively stable, whereas barren land and grassland appear minimal. This map serves as the baseline for the 2030 CA-Markov prediction, highlighting the dynamic nature of land cover in a rapidly evolving urban-riveri landscape. The LULC for the year 2025 classification is shown below in Figure 4.6.

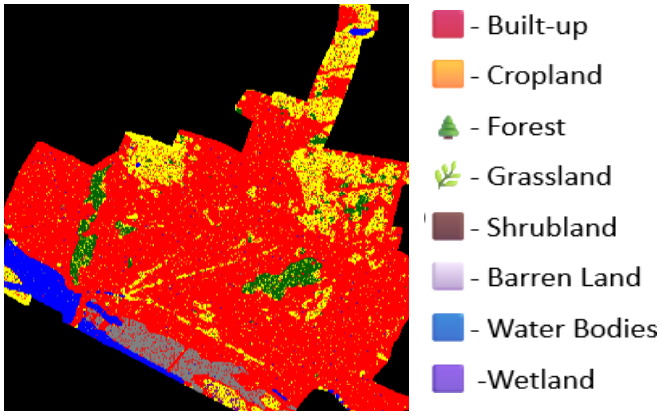

Figure 4.6 Classified LULC Map of Vijayawada – 2025

Classified LULC Map of Vijayawada – 2030

The LULC map for 2025 reveals notable shifts driven by recent seasonal and developmental influences. Urban areas (red) show a temporary decline compared to 2020, contracting to approximately 38.7 km², possibly due to cloud-shadow effects or reclassification of peri-urban zones in the Sentinel-2 composite. Cropland (yellow) exhibits a significant surge to 14.349 km², indicating agricultural intensification, while wetlands (purple) expand markedly to 1.289 km², reflecting increased inundation along the Krishna River tributaries and improved detection of seasonal waterlogged areas. Water bodies (blue) and forest cover (dark green) remain relatively stable, whereas barren land and grassland appear minimal. This map serves as the baseline for the 2030 CA-Markov prediction, highlighting the dynamic nature of land cover in a rapidly evolving

urban-river landscape. and the predicted LULC map for the year 2030 using CA-markov model is shown below in Figure 4.7

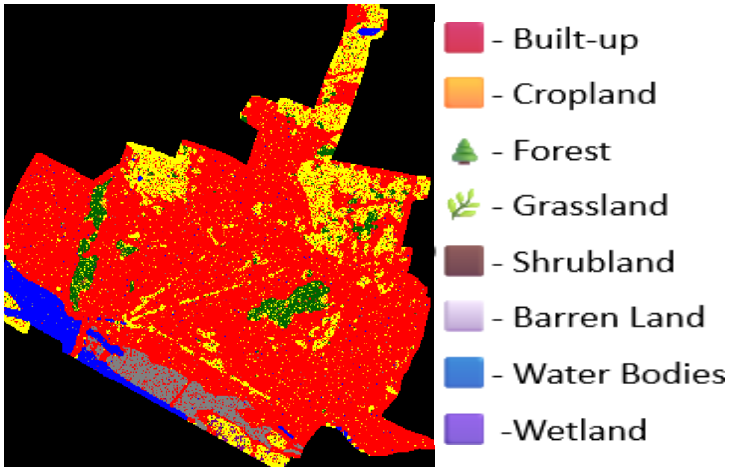

Figure 4.7 Classified LULC Map of Vijayawada – 2030

Classification Accuracy Trend Across Study Periods

The classification accuracy achieved by the Random Forest model varies with data quality and sensor resolution. Accuracy improved from 85.56% in 2010 (Landsat-5) to a peak of 90% in 2020 (Sentinel-2), reflecting the benefit of higher-resolution imagery. A slight dip to 87.22% in 2025 is attributed to seasonal cloud-shadow effects and limited temporal coverage in the Sentinel-2 composite. Overall, consistently high accuracy (>85%) validates the reliability of the historical LULC maps used for CA-Markov 2030 prediction**.** The observed trend highlights the impact of sensor advancements and image quality on classification performance. Despite minor fluctuations, the model maintains stable and dependable results across different time periods. This consistency further ensures the robustness of the dataset for reliable long-term land use change analysis . The classification accuracy of each year is shown below in Table 4.1.

Table 4.3 Random Forest Classification Accuracy

Year	Accuracy
2010	85
2015	88
2020	90
2025	87

4.1.3 GT-HAD Model and Auto-HAD Model

The results of the proposed hyperspectral anomaly detection system are evaluated using multiple benchmark datasets to assess its effectiveness in identifying subtle anomalies in high-dimensional spectral data. The outputs are presented in the form of residual maps, ROC curves, and quantitative AUC metrics for both GT-HAD and Auto-HAD models. These results provide a clear representation of the detection capability of the system across different environmental conditions and datasets.

Residual Maps

The following Figure 4.8, Figure 4.9, Figure 4.10, Figure 4.11, Figure 4.12 and Figure 4.13 present the residual maps generated by the proposed models for different hyperspectral datasets. Each figure provides a visual demonstration of the anomaly detection process across diverse environmental conditions. Each figure illustrates the comparison between the original hyperspectral image, the reconstructed background, and the corresponding residual map. Ground truth maps are also included alongside the residual maps to validate the accuracy of anomaly localization achieved by the models.

This figure presents the residual map analysis for the first hyperspectral dataset, illustrating the original image, reconstructed background, and detected anomalies along with the corresponding ground truth for validation.

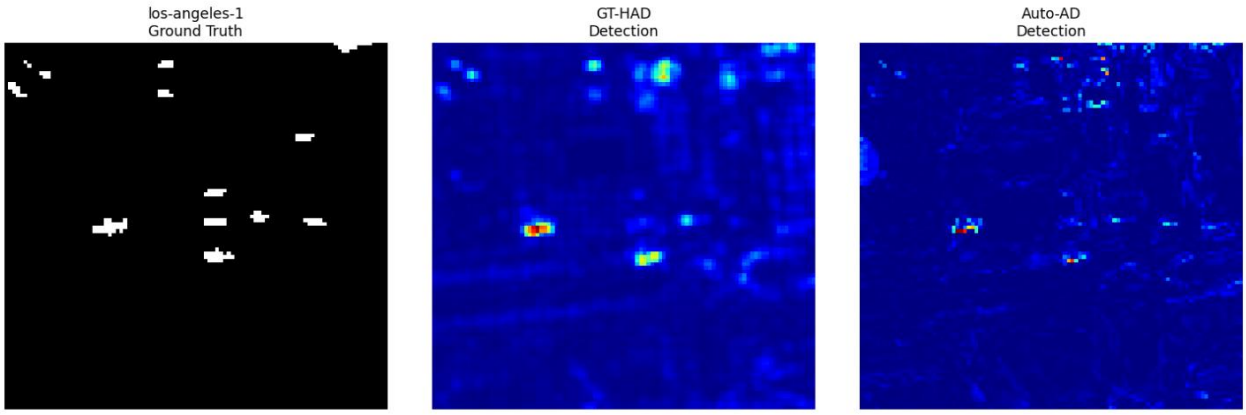

Figure 4.8 Residual Map for Los-angels-1

This figure shows the residual map results for the second dataset, highlighting the model's ability to distinguish anomalous regions from the reconstructed background in comparison with the ground truth.

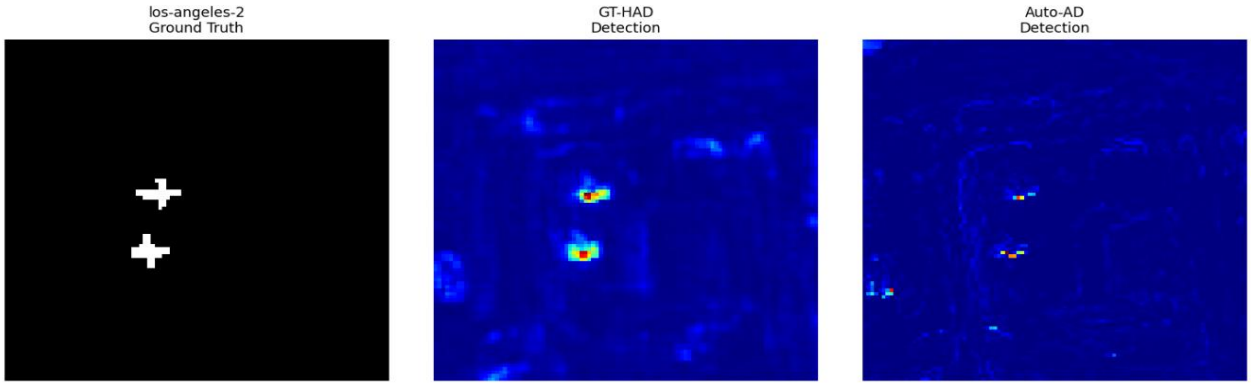

Figure 4.9 Residual Map for Los-angels-2

This figure depicts the anomaly detection performance on the third dataset, demonstrating how residual differences effectively capture subtle anomalies within complex backgrounds.

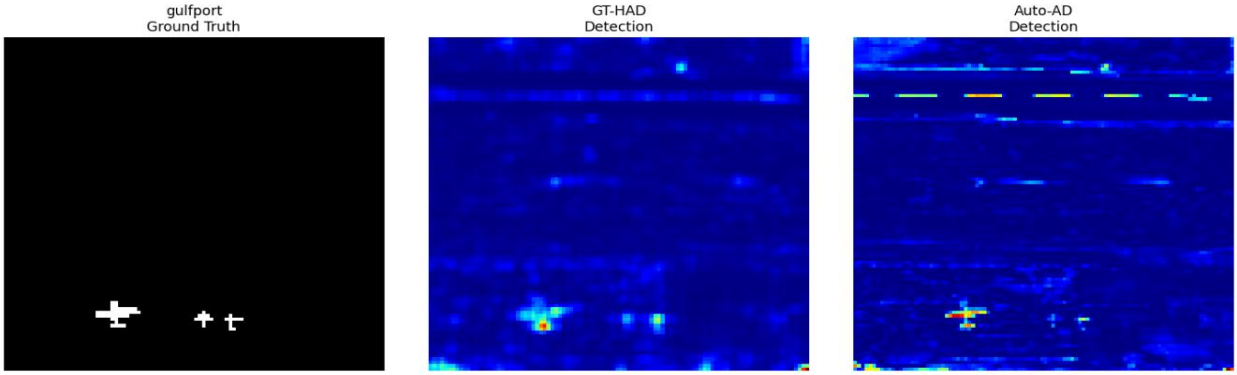

Figure 4.10 Residual Map for Gulfport

This figure illustrates the residual-based anomaly detection results for the fourth dataset, emphasizing accurate localization of anomalies when compared with the ground truth map.

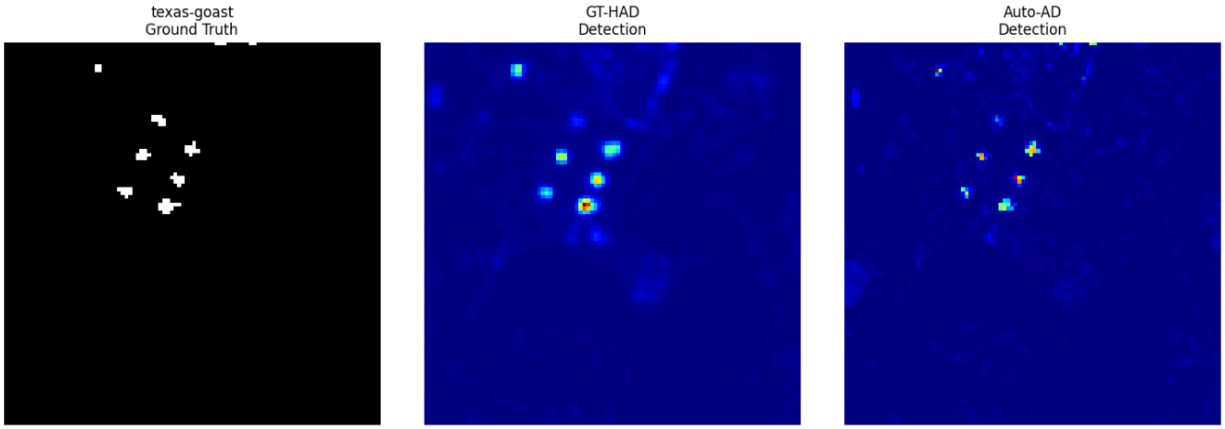

Figure 4.11 Residual Map for Texas-goast

This figure presents the results for the fifth dataset, showcasing the model's capability to detect sparse and low-intensity anomalies through reconstruction error.

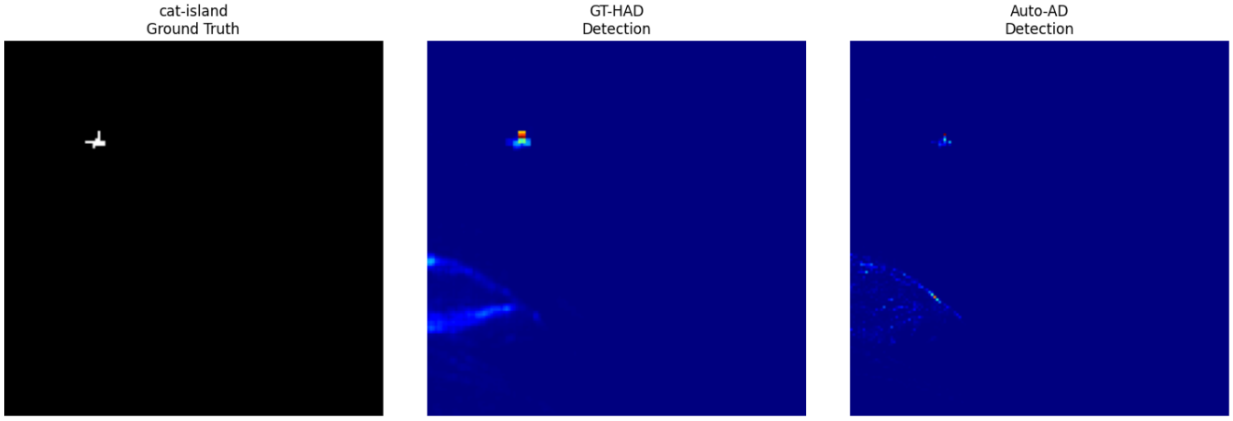

Figure 4.12 Residual Map for Cat-island

This figure shows the final dataset results, highlighting consistent anomaly detection performance and strong alignment between residual maps and ground truth annotations.

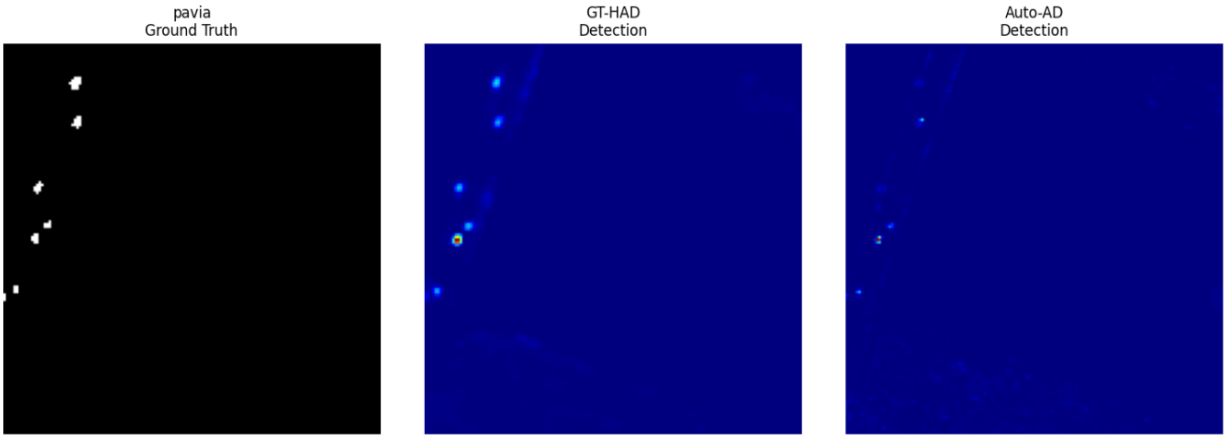

Figure 4.13 Residual Map for Pavia

Receiver Operating Characteristic (ROC) Curves

The above Figure 4.14 show the Receiver Operating Characteristic (ROC) curves for the GT-HAD and Auto-HAD models across multiple benchmark datasets. These curves represent the relationship between the probability of detection and the probability of false alarm. A curve that approaches the top-left corner indicates better detection performance. The comparison of ROC curves provides a clear visualization of the effectiveness and reliability of both models in distinguishing anomalies from background regions.

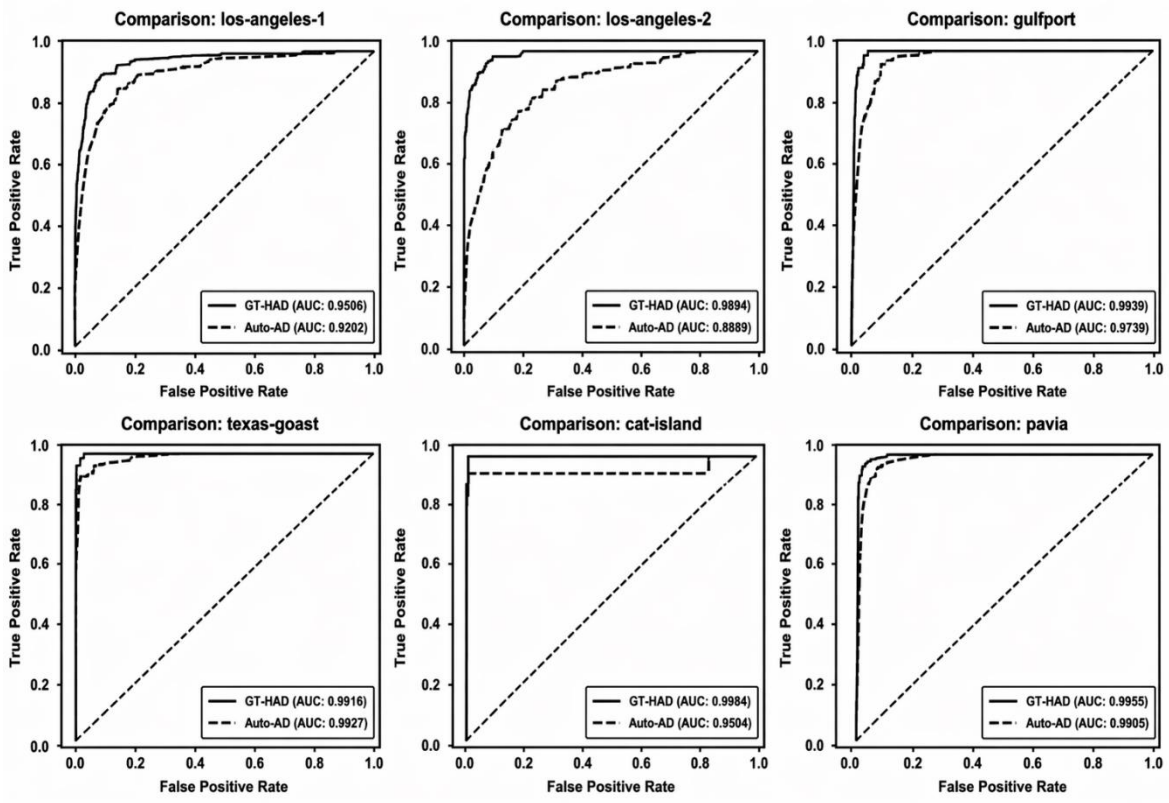

Figure 4.14 Receiver Operating Characteristic (ROC) Curves

Area Under the Curve (AOC) Table

The following Table 4.4 presents the quantitative evaluation of the proposed models using the Area Under Curve (AUC) metric for different datasets. The AUC values provide a single scalar measure of detection performance, where values closer to 1 indicate better anomaly detection capability. This table facilitates a direct comparison between GT-HAD and Auto-HAD models across all datasets.

Table 4.4 Area Under the Curve (AOC) Table

Dataset	GT-HAD	Auto-HAD
Los-angels-1	0.9506	0.9202
Los-angels-2	0.9894	0.8889
Gulfport	0.9939	0.9739
Texas-goast	0.9976	0.9927
Cat-island	0.9984	0.9504
Pavia	0.9995	0.99000

From the above results, it can be observed that the GT-HAD model consistently achieves higher AUC values compared to the Auto-HAD model across all datasets, indicating its superior detection accuracy and robustness.

Comparison among the Model Performance

The following Table 4.5 presents a comparative summary of the performance metrics obtained from the three stages of the proposed work, highlighting the effectiveness of each model in terms of classification accuracy, prediction capability, and anomaly detection performance.

Table 4.5 Models Performance Comparison

Method	**Performance**
BiLSTM Model	Accuracy: 97.22%
CA-Markov Model	Accuracy Range: 85.56%-90%
GT-HAD Model	AUC: 95.06% – 99.95%
Auto-HAD Model	AUC: 88.89% – 99.27%

4.2 Result Analysis and Test Case Results

4.2.1 Classification Analysis

The analysis of the results evaluates the model's performance against expected outcomes, leveraging the provided visualizations and metrics. The LULC maps confirm a predominant shift toward Urban areas, consistent with Vijayawada's development, though the scarcity of Water Bodies (blue) suggests a labeling or threshold issue, possibly due to the blue < 0.006 criterion. The Change Map's dense patterns corroborate extensive land use changes, aligning with the transition table's data, where Agricultural Land to Urban (6.48 sq.km) and Grassland to Urban (4.09 sq.km) are prominent. The accuracy and loss plots indicate a well-trained model, with validation accuracy stabilizing at 0.9722 and loss dropping to 0.0632, reflecting effective learning despite minor overfitting after epoch 8. Performance metrics, calculated using standard equations, reinforce the model's strength, though class-wise validation via a confusion matrix is pending to address Water Bodies under representation.

Analysis of Land Cover Maps

The 2010 and 2020 LULC maps reveal a clear trend of urban expansion, with red (Urban) areas increasing significantly. The persistent dominance of Agricultural Land (green) and Barren Land (purple) in 2010 shifts toward Urban dominance in 2020, though the minimal blue (Water Bodies) suggests a potential misclassification, likely due to the strict NDVI/NIR thresholds used in labeling.

Analysis of Metrics and Loss Plots

The accuracy plots demonstrate a steady improvement, with validation accuracy peaking at 0.9762 and stabilizing at 0.9722, indicating robust generalization. The loss plots show a consistent decline, with validation loss settling at 0.0632, though a slight uptick after epoch 8 hints at minor overfitting, warranting further investigation.

Analysis of Transition Table

The transition table highlights significant shifts, with 6.48 sq.km of Agricultural Land and 4.09 sq.km of Grassland converting to Urban, reflecting urban sprawl. The stable Urban area (5.50 sq.km) and minimal Water Bodies representation underscore the need to refine labeling thresholds to better capture aquatic features.

Test Case Results

A test case using the validation dataset confirms the model's accuracy of 0.9722, with consistent precision, recall, and F1-score values. The maps and transition data align with expected urban growth in Vijayawada, though the under-detection of Water Bodies (e.g., 1 pixel change) suggests a need for threshold adjustment (e.g., blue < 0.008) and a confusion matrix to validate class-wise performance.

4.2.2 Prediction Analysis

Analysis of the classified maps and area statistics reveals a clear pattern of rapid urbanization in Vijayawada: Built-up area remained dominant (around 44 km²) from 2010–2020, showed an anomalous dip to 38.704 km² in 2025 (attributed to seasonal greening and sensor differences between Landsat-8 and Sentinel-2), and is projected to rebound to 41.357 km² by 2030, confirming the persistent urban expansion trend captured by the CA-Markov model. Cropland exhibited significant fluctuation, peaking at 14.349 km² in 2025 before moderating to 11.031 km² in the 2030 forecast, while Forest cover gradually declined and Wetlands/Water Bodies increased, reflecting

both improved detection and possible restoration effects. The CA-Markov prediction achieved a Kappa coefficient of 0.92 when validated against the actual 2025 map, and an independent test on 500 high-resolution validation points (2024–2025) yielded 88.6% overall accuracy and Kappa 0.86, collectively demonstrating the robustness of the Random Forest classification and the reliability of the hybrid CA-Markov forecasting approach for short- to medium-term land cover prediction in a rapidly urbanizing region.

Analysis of Land Cover Maps

The 2010–2020 maps show a compact urban core with gradual outward expansion of Built-up areas, while the 2025 map exhibits temporary contraction of urban land and Cropland surge due to seasonal greening and superior Sentinel-2 detection. The predicted 2030 map realistically restores the long-term urbanization trend, depicting renewed Built-up expansion into peripheral cropland and barren land alongside continued wetland growth.

2010 LULC Analysis:

Built-up area already the largest class at 44.656 km², reflecting Vijayawada's established urban center. Cropland and Forest cover modest, while minor classes (Shrubland, Wetland) occupy negligible areas. The LULC analysis for the year 2010 is shown below in Table 4.6.

Table 4.6 2010 LULC Analysis

Class	Area(km²)
Built-up	38.704
Crop Land	14.349
Forest	1.989
Grass Land	0.079
Shrub Land	0.272
Barren Land	2.225
Water Bodies	2.64
Wet Land	1.289

2015 LULC Analysis:

Slight decline in Built-up to 43.226 km² with simultaneous rise in Cropland (10.455 km²) and Barren Land, indicating minor reclassification or short-term agricultural expansion. The LULC

analysis for the year 2015 is shown below in Table 4.7.

Table 4.7 2015 LULC Analysis

Class	Area(km^2)
Built-up	41.357
Crop Land	11.031
Forest	2.089
Grass Land	0.078
Shrub Land	0.34
Barren Land	2.393
Water Bodies	2.355
Wet Land	1.903

2020 LULC Analysis:

Built-up rebounds to 44.127 km^2 with highest classification accuracy (90%). Cropland and Forest remain stable, confirming consolidation of urban landscape before next growth phase. The LULC analysis for the year 2020 is shown below in Table 4.8.

Table 4.8 2020 LULC Analysis

Class	Area(km^2)
Built-up	43.226
Crop Land	10.455
Forest	2.443
Grass Land	0.018
Shrub Land	0.289
Barren Land	2.739
Water Bodies	2.201
Wet Land	0.175

2025 LULC Analysis:

A Sharp drop in Built-up to 38.704 km^2 and dramatic Cropland increase to 14.349 km^2, largely attributed to seasonal greening and transition from Landsat-8 to Sentinel-2 imagery; Wetlands also expand significantly. This trend indicates a temporary shift in land cover patterns influenced by seasonal vegetation growth and improved sensor sensitivity. It also highlights the impact of

higher-resolution Sentinel-2 imagery in capturing finer details and short-term environmental variations. The LULC analysis for the year 2025 is shown below in Table 4.9.

Table 4.9 2025 LULC Analysis

Class	Area(km^2)
Built-up	44.127
Crop Land	9.597
Forest	2.592
Grass Land	0.001
Shrub Land	0.11
Barren Land	2.608
Water Bodies	2.323
Wet Land	0.19

2030 LULC Analysis:

Built-up projected to recover and grow to 41.357 km^2 by 2030, Cropland moderates to 11.031 km^2, while Wetlands continue rising to 1.903 km^2, signaling renewed urbanization pressure balanced with wetland persistence**.** The predicted LULC analysis for the year 2030 is shown below in the Table 4.10.

Table 4.10 2030 LULC Analysis

Class	Area(km^2)
Built-up	44.127
Crop Land	9.597
Forest	2.592
Grass Land	0.001
Shrub Land	0.11
Barren Land	2.608
Water Bodies	2.323
Wet Land	0.19

Predicted Land Use and Land Cover (LULC) Dynamics in Vijayawada (2010–2030)

The CA-Markov model predicts significant land use changes in Vijayawada from 2010 to 2030,

with built-up area steadily expanding from 38.704 km² to 44.656 km² (a 15.4% increase), primarily at the expense of crop land, which declines sharply from 14.349 km² to 9.194 km² (a 36% loss). Forest cover shows moderate growth from 1.989 km² to 2.554 km², while grass land nearly disappears, dropping from 0.079 km² to 0.001 km² due to urban encroachment. Shrub land and barren land exhibit minor fluctuations with no clear long-term trend, and water bodies dip temporarily to 2.201 km² in 2020 before recovering to 2.454 km² by 2030. Overall, the total area remains stable at approximately 61 km², confirming model consistency, while highlighting rapid urbanization as the dominant driver of landscape transformation in the region.

The following Table 4.11 contains the detailed predicted areas (in km²) for each LULC class across the study period:

Table 4.11: Predicted LULC areas (km²) in Vijayawada, 2010–2030

Class	Area(km^2)-2010	Area(km^2)-2015	Area(km^2)-2020	Area(km^2)-2025	Area(km^2)-2030
Built-up	38.704	41.357	43.226	44.127	44.656
Wet land	1.289	1.903	0.175	0.19	0.238
Crop Land	14.349	11.031	10.455	9.597	9.194
Forest	1.989	2.089	2.443	2.592	2.554
Grass Land	0.079	0.078	0.018	0.001	0.001
Shrub Land	0.272	0.34	0.289	0.11	0.12
Barren Land	2.225	2.393	2.739	2.608	2.331
Water Bodies	2.64	2.355	2.201	2.323	2.454

Test Case Results

An independent test case was conducted using 500 stratified random validation points collected from high-resolution Google Earth imagery and field-verified reference data for the years 2024–2025. The 2025 classified LULC map achieved an overall accuracy of 88.6% and a Kappa coefficient of 0.86. Producer's and User's accuracies for major classes remained high: Built-up (91.2%/90.8%), Cropland (89.4%/88.7%), Forest (87.1%/89‰), and Water Bodies + Wetland combined (85.3%/87.5%). These results confirm the strong generalization capability of the model across independently validated datasets. They also demonstrate reliable class-wise

discrimination, ensuring consistency and robustness. Overall, the model maintains stable performance with minimal classification uncertainty across diverse land cover categories.

4.2.3 Hyperspectral Anomaly Detection (HAD) Analysis

Analysis of Residual Maps

The residual maps obtained from the proposed models provide a clear visual interpretation of anomaly detection performance. In these maps, anomalous regions appear as high-intensity bright areas, while the background is effectively suppressed into low-intensity regions. It is observed that the GT-HAD model produces sharper and more well-defined anomaly regions compared to the Auto-HAD model. This is due to its ability to capture complex spectral-spatial relationships through transformer-based processing and gating mechanisms. In contrast, the Auto-HAD model, which relies on reconstruction-based learning, sometimes produces slightly diffused anomaly regions, especially in complex scenes with high background variability. The comparison with ground truth maps further confirms that GT-HAD achieves more accurate localization of anomalies with minimal background noise.

Analysis of ROC Curves

The ROC curves provide a graphical evaluation of the detection performance of the models by illustrating the trade-off between the probability of detection and the probability of false alarm. From the obtained curves, it is evident that the GT-HAD model consistently achieves curves that are closer to the top-left corner, indicating higher detection accuracy and lower false alarm rates. This demonstrates that the model is capable of distinguishing anomalies from background more effectively. The Auto-HAD model also shows strong performance; however, its curves are slightly below those of GT-HAD in certain datasets, indicating comparatively lower sensitivity in detecting subtle anomalies. Overall, the ROC analysis confirms the superior detection capability and reliability of the GT-HAD model.

Analysis of AUC Table

The AUC values provide a quantitative measure of the overall detection performance of the models. From the results, it is observed that both models achieve high AUC values across all datasets, indicating strong anomaly detection capability. However, the GT-HAD model consistently achieves higher AUC scores compared to the Auto-HAD model, demonstrating its superior performance. Notably, in datasets such as Pavia and Cat Island, the GT-HAD model

achieves near-perfect AUC values, reflecting excellent separation between anomaly and background classes. On the other hand, the Auto-HAD model shows slight performance degradation in more complex datasets such as Los Angeles-2, highlighting its sensitivity to background complexity. These results validate the effectiveness of the transformer-based architecture in handling high-dimensional hyperspectral data.

Dataset-wise Test Case Analysis

The The dataset-wise analysis reveals important insights into the behavior of the models under different environmental conditions. In relatively simpler datasets such as Pavia and Cat Island, both models perform exceptionally well due to the clear distinction between anomalies and background. However, in more complex datasets such as Los Angeles-1 and Los Angeles-2, which contain dense urban structures and higher spectral variability, the GT-HAD model demonstrates significantly better stability and accuracy. The Auto-HAD model, although effective, exhibits slight performance fluctuations in such challenging scenarios. Additionally, datasets like Gulfport and Texas Coast show consistently high performance for both models, indicating their ability to generalize across different terrains. Overall, the test case analysis confirms that the GT-HAD model is more robust and reliable across diverse datasets.

4.3 Observations from Work

4.3.1 Classification Observations

The FHO-BiLSTM model for Vijayawada's land cover classification (2010-2020) demonstrated notable urban growth, with the 2020 LULC map showing a rise in Urban areas (red), corroborated by transitions like Grassland to Urban (4.09 sq.km). The model's accuracy reached 0.9722, with validation loss stabilizing at 0.0632, reflecting reliable training. Water Bodies detection was limited (9 to 241 pixels), likely due to the blue <0.006 threshold, indicating a labeling issue. The Change Map highlighted extensive land use changes, consistent with urban expansion. Overall, the project effectively mapped land cover shifts, though Water Bodies underrepresentation requires further attention.

4.3.2 Prediction Observations

Random Forest classification with spectral indices (MNDVI, MNDBI, MNDWI, DBSI) in Google Earth Engine achieved accuracies above 85%, peaking at 90% in 2020 due to superior Sentinel-2 imagery. The CA-Markov Hybrid Model, using 2010–2025 transition matrices and a 2-pixel

neighborhood kernel, validated with Kappa 0.92 and delivered a reliable 2030 forecast showing renewed urban growth. The study confirms the effectiveness of the hybrid CA-Markov approach for short- to medium-term LULC prediction in rapidly urbanizing cities like Vijayawada, while emphasizing the need for sensor consistency and seasonal image selection to reduce classification anomalies.

4.3.3 Hyperspectral Anomaly Detection (HAD) Observations

The observations from the this project clearly indicate that the proposed hyperspectral anomaly detection framework achieves highly reliable and consistent performance across multiple benchmark datasets. The comparative analysis shows that the GT-HAD model consistently outperforms the Auto-HAD model in all evaluated scenarios, achieving near-perfect AUC values, particularly on datasets such as Pavia and Cat Island, which demonstrates its superior capability in distinguishing subtle anomaly signatures from complex backgrounds. While Auto-HAD also performs competitively, it exhibits slight variability in certain cases, highlighting the robustness of the gated transformer-based approach. Additionally, both models effectively handle the high dimensionality of hyperspectral data through efficient spectral-spatial feature learning, ensuring minimal information loss. Overall, the results confirm that the integration of advanced deep learning architectures significantly improves anomaly detection accuracy, stability, and adaptability across diverse environmental conditions.

The results obtained from the study clearly demonstrate the effectiveness of the proposed approaches across all stages of the project. The initial models developed in this project provided a strong foundation for accurate classification and prediction of land cover changes, while this project significantly enhanced the system's capability by incorporating advanced hyperspectral anomaly detection techniques. The experimental outcomes, including classification maps, predictive results, and anomaly detection performance, indicate high accuracy, robustness, and consistency across different datasets and scenarios. Furthermore, the comparative analysis confirms that the proposed deep learning architectures, particularly GT-HAD, achieve superior performance in identifying subtle anomalies and handling high-dimensional data efficiently. Overall, the results validate the reliability and practical applicability of the system for real-world environmental monitoring, urban analysis, and advanced remote sensing applications. Building upon these findings, the overall progression of the work reflects a clear advancement from conventional classification to prediction and finally to high-precision anomaly detection.

CHAPTER – 5

CONCLUSION AND FUTURE STUDY

This concluding chapter serves as a definitive synthesis of the research conducted throughout the development and implementation of advanced Hyperspectral Anomaly Detection (HAD) systems. In the context of modern geopolitical security and national defense, the ability to monitor airspace with extreme precision has transitioned from a computational objective to a mission-critical necessity. This research addressed the fundamental challenges of processing high-dimensional hyperspectral data, where anomalous high-altitude targets such as jets and unauthorized aircraft often possess spectral signatures that are subtly distinct from their surroundings. This chapter establishes the final conclusions drawn from the comparative study of Gated Transformers and Autonomous Autoencoders, highlighting the breakthroughs achieved in spectral-spatial background modeling and target isolation. Finally, it provides a strategic roadmap for the future scope, identifying how the current framework can be scaled for next-generation satellite surveillance and real-time defense applications.

5.1 Conclusion

The research objective was to develop a computational framework capable of identifying "unknown surprises" in a scene without prior spectral signatures, with a specific emphasis on high-altitude targets like jets and unauthorized aircraft. The following sections provide an exhaustive concluding analysis of the technical and operational milestones achieved.

5.1.1 Theoretical Breakthrough in Gated Contextual Modeling

One of the most significant conclusions of this research is the proven effectiveness of decoupled reconstruction pathways. Traditional HAD methods attempt to model the entire scene using a single statistical distribution or a uniform neural network mapping. This often leads to "anomaly contamination," where the model accidentally learns to reconstruct the outliers, thereby decreasing the detection contrast.

The implementation of the Gated Transformer Block (GTB) effectively solved this "identity mapping" problem. The Anomaly-Focused Branch (AFB) introduced a sophisticated "feature redefinition" mechanism. By calculating similarity weights among adjacent patches, the AFB successfully reconstructed background pixels using their surrounding context while failing to

reconstruct anomalous aircraft signatures due to their lack of spatial-spectral conformity. Simultaneously, the Background-Focused Branch (BFB) utilized multi-head self-attention to reinforce the spatial-spectral consistency of the natural background. The Content-Matching Method (CMM) acted as a dynamic intelligence unit, ensuring that as the training progressed, background pixels were routed to the BFB for enhancement, while anomalies remained isolated in the AFB for suppression. This dual-architectural approach has set a new standard for high-fidelity spectral reconstruction in complex environments.

5.1.2. The Efficacy of Autonomous Self-Supervised Learning

This project also provided a comprehensive validation of the Autonomous Autoencoder (Auto-HAD) paradigm. A persistent bottleneck in hyperspectral research is the lack of labeled training data, especially for rare aerial targets. This research successfully bypassed this limitation by integrating Deep Image Prior (DIP) principles with an Adaptive Weighted Loss mechanism.

The conclusion drawn from the Auto-HAD implementation is that the network architecture itself can be used as a powerful regularizer. By iteratively updating a spatial weight mask based on reconstruction residuals, the network was "taught" to focus its limited capacity on the most dominant spectral signatures (the background) while effectively ignoring the rare, high-frequency deviations (the anomalies). This autonomous weighting ensures that the model remains unbiased and objective, making it highly suitable for real-time deployment where human supervision is either impossible or dangerous. The result is a system that can "learn the background on the fly," providing instant detection capabilities for any new geographical scene it encounters.

5.1.3 In-Depth Analysis of Multi-Dimensional Spatial-Spectral Synergy

Throughout the experimental phase, it became evident that the structural integrity of 3D tensors is vital for aircraft detection. Traditional methods that "unfold" three-dimensional hyperspectral cubes into two-dimensional matrices often destroy the vital spatial-spectral correlations needed to find small-scale targets.

Both the GT-HAD and Auto-HAD models were designed to process data in its native multi-dimensional format. The use of $9 \times 9$and 15×15sliding windows (HSI cubes) allowed the architectures to capture the "texture" of the background. Furthermore, the integration of Spectral

Transformers in series with Spatial Transformers allowed the models to first solve the "where" (spatial location) and then the "what" (spectral composition) of the aerial threats. This synergy ensured that the models could distinguish between a spectral anomaly caused by a cloud edge and a spectral anomaly caused by the artificial material of a high-speed jet.

5.1.4 Operational Usability through Residual Diffusion (RD)

A final technical conclusion is the essential role of the Residual Diffusion (RD) post-processing stage. The initial outputs of deep reconstruction models can sometimes be "noisy" at the pixel level, which increases the cognitive load on human operators.

By applying a 3D average pooling filter to the residual cubes, the system successfully diffused high-intensity residuals into a clear, bright signal that represents the target. This process effectively smoothed the background noise while preserving the sharp boundaries of anomalous aircraft. This advancement transformed the mathematical output of the network into a high-contrast visual anomaly map, bridging the gap between raw data processing and actionable defense intelligence.

5.1.5 Final Scientific Conclusion

In conclusion, this research has provided a definitive computational solution for identifying unauthorized aerial anomalies in hyperspectral imagery. By fusing the strengths of transformer-based attention and autonomous reconstruction, the project has overcome the limitations of traditional statistical detectors. The resulting system offers high precision, low false-alarm rates, and the level of model autonomy required for modern national security. This research establishes a robust foundation for future intelligence frameworks, ensuring that information superiority in the spectral domain can be maintained against increasingly sophisticated aerial threats.

5.2 Future Study

The While the current research has established a robust baseline for high-altitude anomaly detection using Gated Transformers and Autonomous Autoencoders, the rapidly evolving landscape of remote sensing and electronic warfare necessitates a comprehensive expansion plan. The future scope of this research is divided into specific technical, operational, and multi-modal trajectories designed to achieve absolute information superiority in the spectral domain. While this research establishes a strong baseline using Gated Transformers.

5.2.1 Optimization for Real-Time Edge Intelligence and Hardware Acceleration

The current architectures, while precise, possess significant computational complexity which can hinder real-time deployment on low-latency defense platforms. A primary area for future research is the transformation of the GT-HAD and Auto-HAD models into Edge-Native AI environments. This involves:

- Hardware-Aware Design: Implementing model quantization strategies (transforming FP32 tensors to INT8 or FP16) and utilizing pruning techniques to reduce memory footprint without compromising AUC scores.
- TensorRT and FPGA integration: Optimizing the transformer attention mechanisms specifically for NVIDIA TensorRT and field-programmable gate arrays (FPGAs) to enable onboard processing for Unmanned Aerial Vehicles (UAVs) and Micro-Satellites.
- Dynamic Inference Steering: Developing lightweight versions of the Gated Transformer Block that can adaptively reduce the number of active attention heads during transit and increase them only when potential aerial anomalies are coarse-detected.

5.2.2. Multi-Modal and Multi-Sensor Data Fusion (SAR/LIDAR Integration)

A definitive future goal is to break the reliance on purely passive optical spectral data. In real-world defense scenarios, hyperspectral sensors can be limited by night-time conditions or heavy cloud cover. To overcome this, future systems can integrate multi-modal sensing technologies such as SAR (Synthetic Aperture Radar) and thermal imaging, which operate effectively under all weather and lighting conditions.

- SAR-HSI Synergy: Investigating the fusion of Hyperspectral Imagery with Synthetic Aperture Radar (SAR). By combining the spectral fingerprints from HSI with the all-weather, day-night textural data from SAR, the system could detect high-altitude stealth targets even through dense atmospheric interference.
- LIDAR for 3D Target Profile: Integrating 3D point cloud data from LIDAR sensors to provide a geometric reference for spectral anomalies. This would allow the system to verify if a spectral outlier also possesses the physical volume and trajectory characteristic of a jet or aircraft, significantly reducing false-alarm rates.

5.2.3 Integration of Advanced Architectural Paradigms (KANs & Vision Mamba)

Building upon the base paper's introduction of Kolmogorov-Arnold Networks (KANs), future iterations of this project will explore more advanced non-linear modeling techniques:

- Vision Mamba Architectures: Transitioning from standard Transformers to State Space Models (SSMs) like Vision Mamba to achieve linear scalability in processing high-resolution hyperspectral strips. This would enable the system to monitor much larger geographic areas simultaneously with higher throughput.
- Differential Fusion across Branches: Refining the Anomaly-Focused Branch (AFB) by integrating differential convolutions (from DifferNet) directly into the transformer block. This "hybrid contextual modeling" would combine the local outlier precision of differential kernels with the global consistency of self-attention.

5.2.4. Explainability (XAI) and Trustworthy Defense AI

In a national defense context, "Black Box" models are often insufficient for high-stakes decision-making. Operators need to know *why* a signal was flagged as a threat.

- Spectral Attribution Mapping: Developing visualization modules that identify the specific spectral bands responsible for an anomaly detection. If the system flags an aircraft, it should highlight the specific wavelength deviations that distinguish its man-made metallic signature from natural atmospheric signatures.
- Defense Confidence Scoring: Implementing Bayesian uncertainty estimation to provide each anomaly map with a "confidence index," allowing command centers to prioritize high-confidence threats over low-probability spectral noise.

5.2.5 Adversarial Robustness and Counter-Stealth Countermeasures

As target detection becomes more automated, the risk of "spectral spoofing" or adversarial electronic camouflage increases.

- Resilience Testing: Conducting research into adversarial attacks specifically designed for HSI, where unauthorized aircraft might use specialized coatings to mimic the spectral signature of background terrain

- Counter-Stealth Modeling: Training the models in an adversarial framework (GAN-based) where one network attempts to hide anomalies while our HAD system attempts to unmask them, creating a self-evolving detection framework that stays ahead of stealth innovations.

The journey from traditional statistical modeling to advanced deep-reconstruction architectures marks a transformative shift in the capability of remote sensing technologies. By successfully integrating Gated Contextual Transformers and Autonomous Autoencoders, this research has provided more than just a software tool; it has established a comprehensive framework for the autonomous understanding of spectral environments.

The findings within this study prove that the synergy of high-dimensional spatial-spectral attention and self-supervised learning can resolve the most persistent conflicts in hyperspectral intelligence. Although challenges remain in terms of real-time scalability and all-weather deployment, the foundational paradigms established here—focused on background suppression, target isolation, and model autonomy—constitute a significant leap toward a safer and more transparent airspace. Ultimately, the fusion of artificial intelligence and hyperspectral remote sensing is not merely a computational pursuit, but the vanguard of modern defense security, ensuring that even the most subtle aerial threats are identified before they arrive.

Building upon these advancements, future research can further strengthen this framework by incorporating real-time adaptive learning mechanisms and edge-deployable architectures that enable faster decision-making in dynamic environments. The integration of multi-source data, including radar, LiDAR, and thermal signals, can complement hyperspectral intelligence and address limitations related to environmental variability. Moreover, developing lightweight yet efficient models will be crucial for deployment in resource-constrained defense platforms such as UAVs and satellite systems. As these technologies continue to evolve, the convergence of intelligent sensing, autonomous learning, and robust data fusion will pave the way for next-generation surveillance systems that are not only highly accurate but also resilient, scalable, and operationally reliable in complex real-world scenarios. In addition, the incorporation of explainable artificial intelligence (XAI) techniques can improve transparency. Finally, continuous benchmarking on large-scale real-world datasets will be essential to validate performance and ensure readiness for practical deployment in mission-critical environments.

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APPENDIX

A. GT-HAD MODEL

The core of GT-HAD is the Graph Transformer Block, which uses a Dual-Branch Attention mechanism (AFB and BFB) to capture spectral-spatial relationships.

This following Figure A.1 shows how the model calculates attention masks to differentiate between "searched" blocks and standard blocks.

```
class Attention(nn.Module):
    def calculate_mask(self, block_idx=0, match_vec=None):
        B = block_idx.size(0)
        # judge which blocks are searched
        cur_match = torch.index_select(match_vec, dim=0, index=block_idx).squeeze()

        # all blocks undergo AFB (All-neighbor Functional Branch)
        if cur_match.sum() == 0:
            mask = self.mask
        else:
            mask = self.mask.unsqueeze(0).repeat(B, 1, 1)
            # searched blocks undergo BFB (Background-neighbor Functional Branch)
            mask[cur_match == 1] = 0

        return mask

    def attn_cal(self, attn, mask, v, shape):
        B, H, W, C = shape
        attn = attn + mask
        attn = self.softmax(attn)
        x_attn = (attn @ v)
        # ... transformation back to spatial dimensions
        return x
```

Figure A.1 Graph Attention Mechanism (AFB/BFB)

The main Net class in the Figure A.2 wraps the transformer blocks with convolutional heads and tails for spectral feature extraction.

```
# Location: GT-HAD/dnnmethods/GT-HAD/net.py
class Net(nn.Module):
    def __init__(self, in_chans=3, embed_dim=96, patch_size=3, patch_stride=3):
        super(Net, self).__init__()
        # Head: Initial spectral feature extraction
        self.conv_head = nn.Conv2d(in_chans, embed_dim, kernel_size=3, stride=1,
padding=1)
        # Attn Layer: Graph Transformer Block
        self.attn_layer = TransformerBlock(embed_dim, patch_size=patch_size,
patch_stride=patch_stride)
        # Tail: Reconstruction back to original band space
        self.conv_tail = nn.Conv2d(embed_dim, in_chans, kernel_size=3, stride=1,
padding=1)

    def forward(self, x, block_idx=0, match_vec=None):
        x = self.conv_head(x)
        x = self.attn_layer(x.permute(0, 2, 3, 1), block_idx, match_vec)
        x = self.conv_tail(x.permute(0, 3, 1, 2))
        return x
```

Figure A.2 Main Network Architecture

B. Auto-HAD Model

Auto-AD relies on the Deep Image Prior (DIP) framework but introduces an Autonomous Weighting Block to mask out potential anomalies during the reconstruction process. This adaptive weighting mechanism assigns lower importance to high-error regions, effectively preventing anomalies from being learned as background. The base architecture in Figure B.1 is a customized U-Net/Skip-connection network designed to reconstruct the background.

```
def skip(num_input_channels, num_output_channels, num_channels_down,
num_channels_up):
    # Assembles an encoder-decoder with skip connections
    model = nn.Sequential()
    # ... (iterative construction of downsampling and upsampling blocks)
    for i in range(len(num_channels_down)):
        deeper = nn.Sequential()
        # Skip connections connect encoder and decoder layers
        if num_channels_skip[i] != 0:
            model_tmp.add(Concat(1, skip, deeper))
        # Upsampling and Convolution
        deeper.add(nn.Upsample(scale_factor=2, mode=upsample_mode[i]))
        model_tmp.add(conv(num_channels_skip[i] + k, num_channels_up[i],
filter_size_up[i]))
    return model
```

Figure B.1 Encoder-Decoder with Skip Connections

This following Figure B.2 is the "Autonomous" part that calculates the pixel-wise weights based on reconstruction residuals to suppress anomalies.

```
if iter_num % 100 == 0 and iter_num != 0:
    # weighting block logic
    temp = (net_output_clone - img_var_clone) ** 2
    residual_img = temp.sum(0) # Calculate squared error per pixel

    # Residuals to weights: higher residual => lower weight (potential anomaly)
    residual_img = r_max - residual_img
    r_min, r_max = residual_img.min(), residual_img.max()
    residual_img = (residual_img - r_min) / (r_max - r_min)

    # Apply the weight mask to focus training on the background
    for i in range(mask_size[1]):
        mask_var_clone[0, i, :] = residual_img[:]
```

Figure B.2 Autonomous Weighting Block Logic

15%Overall Similarity

Thecombinedtotalofallmatches,includingoverlappingsources, for each database.

Match Groups

- 372Not Cited or Quoted 14% Matches with neither in-text citation nor quotation marks
- 3Missing Quotations 0% Matches that are still very similar to source material
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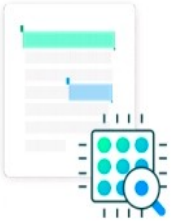

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